Quantum Fourier Transform

Assessment of the QFT algorithm

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key** with the topic tested by every question (page 2), and
2. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count C = number of correct answers and W = number of questions answered wrongly (or with more than one option marked). Section marks $= 3C - W$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy, and CUDA-Q state vectors for the QFT circuits) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- n qubits, $N = 2^n$, $\omega = e^{2\pi i/N}$. A basis state is written $|x\rangle = |x_1 x_2 \dots x_n\rangle$ with $x = \sum_j x_j 2^{n-j}$: **qubit 1 is the most significant bit** and is written first.
- $\mathrm{QFT}\,|x\rangle = \dfrac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy}\,|y\rangle$, $\quad \mathrm{QFT}^{-1}\,|y\rangle = \dfrac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega^{-xy}\,|x\rangle$. Binary fraction: $0.x_l x_{l+1} \dots x_m = \frac{x_l}{2} + \frac{x_{l+1}}{4} + \dots + \frac{x_m}{2^{m-l+1}}$.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_k = \mathrm{diag}\left(1, e^{2\pi i/2^k}\right)$, so $R_1 = Z$, $S = R_2$, $T = R_3$; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- **Textbook QFT circuit:** for $j = 1, \dots, n$: apply H to qubit j, then for $k = 2, \dots, n-j+1$ a controlled-R_k with control qubit $j+k-1$ and target qubit j; finally SWAP qubits j and $n+1-j$ for $j \le n/2$.
- **Phase estimation (QPE):** $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$, $0 \le \varphi < 1$; t counting qubits in $|+\rangle$ control $U^{2^{t-1}}, \dots, U^2, U$; the inverse QFT is applied and the outcome y gives the estimate $\varphi \approx y/2^t$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Mathematics for the QFT

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**A**	Weighted sum of roots of unity	11	**C**	FFT vs. QFT cost (and its limits)
2	**B**	Orthogonality of roots of unity	12	**A**	Cyclotomic product identity
3	**A**	Binary fractions, periodicity of $e^{2\pi i\theta}$	13	**D**	DFT of a vector; Parseval
4	**D**	Unitarity; the 4×4 Fourier matrix	14	**D**	Binary expansion of $2^k\varphi$
5	**A**	Gauss sum, trace of F_4	15	**C**	Algebra of 8th roots of unity
6	**D**	Interference pattern $\lvert\sum e^{ij\theta}\rvert^2$	16	**B**	Properties of F, F^2, $H\otimes H$, CT
7	**B**	Continued fractions, convergents	17	**C**	Root sums when $\gcd(k,N)>1$
8	**C**	Multiplicative order; Shor's gcd step	18	**C**	Determinants of unitary matrices
9	**B**	Parameters of an n-qubit state	19	**A**	Parseval for the un-normalised DFT
10	**B**	Walsh–Hadamard signs	20	**D**	Spectral leakage ($r\nmid N$)

Section B — Quantum Mechanics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**B**	Born rule, partial measurement	31	**D**	Commutation of gates
22	**A**	Global vs. relative phase	32	**B**	State-vector memory cost
23	**A**	Interference with a Hadamard	33	**C**	Bell state in the X basis
24	**B**	Bloch-sphere expectation values	34	**B**	Mixture vs. superposition
25	**C**	Entanglement test	35	**C**	Single-qubit gate actions
26	**D**	Phase kickback (CNOT)	36	**D**	Two-qubit correlations
27	**A**	Controlled-phase matrices	37	**C**	Distinguishability; relative phase
28	**B**	Kickback of controlled-U^{2^j}	38	**D**	Product-state criterion
29	**A**	Post-measurement state	39	**D**	Hadamard test
30	**A**	Unitarity of gates	40	**C**	Repetition and confidence

Section C — QFT Basics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**A**	QFT matrix element ($n=2$)	51	**A**	Constructing the inverse QFT
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43	**C**	Product formula ($n=3$)	53	**B**	QFT vs. NumPy FFT conventions
44	**C**	Controlled-rotation count	54	**D**	QFT $\lvert x\rangle$ is a product state
45	**B**	Two-qubit gates incl. SWAPs	55	**C**	Gate count $n(n-1)/2$
46	**D**	Smallest rotation angle	56	**D**	Two-qubit QFT outputs
47	**A**	Phase on qubit 1; role of SWAPs	57	**A**	Why measuring QFT $\lvert x\rangle$ reveals nothing
48	**C**	QFT^2 = negation	58	**C**	What the QFT does *not* speed up
49	**B**	QFT of the uniform state	59	**B**	Circuit trace: phase build-up
50	**A**	QFT of $\lvert 10\ldots 0\rangle$	60	**D**	Circuit trace: final amplitude, SWAP

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Section D — Advanced QFT

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**D**	QFT of a periodic state $(r \mid N)$	71	**D**	Approximate QFT gate savings
62	**B**	Phases of QFT peaks	72	**A**	Periodic state with $r \nmid N$
63	**D**	Shift theorem	73	**D**	Eigenvalues of the QFT
64	**B**	QPE precision vs. counting qubits	74	**C**	Semiclassical QFT
65	**B**	QPE with exact t-bit phase	75	**A**	Applications of the QFT
66	**C**	The $4/\pi^2$ success bound	76	**C**	Period $\leftrightarrow$ peak spacing
67	**D**	QPE distribution for $\varphi = 1/3$	77	**A**	Approximate QFT error bound
68	**A**	Qubit budget of Shor's algorithm	78	**A**	Non-terminating binary phases
69	**B**	Order finding, $a = 7$, $N = 15$	79	**C**	Shor, $N = 21$: continued fractions
70	**B**	Failure mode $\gcd(k, r) > 1$	80	**C**	Shor, $N = 21$: peak probability

Quick-marking grid

1 **A**	2 **B**	3 **A**	4 **D**	5 **A**	6 **D**	7 **B**	8 **C**	9 **B**	10 **B**
11 **C**	12 **A**	13 **D**	14 **D**	15 **C**	16 **B**	17 **C**	18 **C**	19 **A**	20 **D**
21 **B**	22 **A**	23 **A**	24 **B**	25 **C**	26 **D**	27 **A**	28 **B**	29 **A**	30 **A**
31 **D**	32 **B**	33 **C**	34 **B**	35 **C**	36 **D**	37 **C**	38 **D**	39 **D**	40 **C**
41 **A**	42 **D**	43 **C**	44 **C**	45 **B**	46 **D**	47 **A**	48 **C**	49 **B**	50 **A**
51 **A**	52 **B**	53 **B**	54 **D**	55 **C**	56 **D**	57 **A**	58 **C**	59 **B**	60 **D**
61 **D**	62 **B**	63 **D**	64 **B**	65 **B**	66 **C**	67 **D**	68 **A**	69 **B**	70 **B**
71 **D**	72 **A**	73 **D**	74 **C**	75 **A**	76 **C**	77 **A**	78 **A**	79 **C**	80 **C**

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Answer Bank — questions with detailed solutions

SECTION A · Mathematics for the QFT Q.1–20

Q.1 SINGLE-CORRECT MCQ · Weighted sum of roots of unity

Let $\omega = e^{2\pi i/8}$. The value of $S = \sum_{k=0}^{7} k\,\omega^k$ is

(A) ✓ $\dfrac{8}{\omega - 1}$ **(B)** $\dfrac{8}{1-\omega}$ **(C)** 0 **(D)** $\dfrac{8\omega}{\omega - 1}$

Answer: (A) Detailed solution

What you need to know. Geometric series $\sum_{k=0}^{M-1} z^k = \frac{1-z^M}{1-z}$; differentiating it gives weighted sums. For $\omega = e^{2\pi i/8}$, $\omega^8 = 1$ and $\omega^7 = \omega^{-1}$.

Step-by-step solution. Use $\sum_{k=0}^{M-1} kz^k = \dfrac{z(1 - Mz^{M-1} + (M-1)z^M)}{(1-z)^2}$ (differentiate the geometric series $\sum z^k = \frac{1-z^M}{1-z}$ and multiply by z). With $z = \omega$, $M = 8$, $\omega^8 = 1$ and $\omega^7 = \omega^{-1}$:

$$S = \frac{\omega\,(1 - 8\omega^{-1} + 7)}{(1-\omega)^2} = \frac{8\omega - 8}{(1-\omega)^2} = \frac{-8(1-\omega)}{(1-\omega)^2} = \frac{8}{\omega - 1}.$$

Check: $S \neq 0$ because the weights k break the perfect cancellation of $\sum_k \omega^k = 0$; $\frac{8}{1-\omega}$ has the wrong sign. Numerically $S = -4 + 9.657i$.

Why the other options are wrong.

(B) $\frac{8}{1-\omega}$ is the negative of the correct value: it comes from a sign slip when simplifying $\frac{8\omega-8}{(1-\omega)^2}$ (note $8\omega - 8 = -8(1-\omega)$).

(C) 0 would be right for the *unweighted* sum $\sum_k \omega^k$; the weights k destroy the symmetric cancellation (numerically $S = -4 + 9.66i \neq 0$).

(D) $\frac{8\omega}{\omega-1}$ has an extra factor ω: it forgets that $\omega \cdot \omega^{-1} = 1$ inside $\omega(1 - 8\omega^{-1} + 7)$.

Remember. Weighted root-of-unity sums: differentiate the geometric series, then use $\omega^N = 1$. Always check against the numerical value.

Q.2 SINGLE-CORRECT MCQ · Orthogonality of roots of unity

The number of ordered pairs (x, y) with $x, y \in \{0, 1, \ldots, 15\}$ for which $\sum_{z=0}^{15} e^{2\pi i\,(x+y)z/16} \neq 0$ is

(A) 1 **(B)** ✓ 16 **(C)** 15 **(D)** 31

Answer: (B) Detailed solution

What you need to know. Orthogonality: $\sum_{z=0}^{N-1} \omega^{kz} = N$ if $N \mid k$, otherwise 0.

Step-by-step solution. The sum is a sum of 16th roots of unity: $\sum_{z=0}^{15} \omega^{kz} = 16$ if $16 \mid k$ and 0 otherwise ($\omega = e^{2\pi i/16}$, geometric series). Here $k = x + y \in \{0, \ldots, 30\}$, so we need $x + y \in \{0, 16\}$. $x + y = 0$: one pair $(0, 0)$. $x + y = 16$: $x = 1, \ldots, 15$ with $y = 16 - x$: 15 pairs. Total $1 + 15 = 16$. (Equivalently: $y \equiv -x \pmod{16}$ has exactly one solution y for every x.)

Why the other options are wrong.

(A) 1 counts only $x + y = 0$ and forgets the pairs with $x + y = 16$.

(C) 15 counts only $x + y = 16$ and forgets $(0, 0)$.

(D) 31 treats every sum value $0, \ldots, 30$ as non-zero, i.e. ignores the orthogonality rule.

Remember. For each x there is exactly one $y \equiv -x \pmod N$, so the count is always N.

Q.3 SINGLE-CORRECT MCQ · Binary fractions, periodicity of $e^{2\pi i\theta}$

Let $x = 45$, written with $n = 6$ bits as $x_1x_2\ldots x_6$. The value of $e^{2\pi i\, x/2^4}$ is

(A) ✓ $e^{i\,13\pi/8}$ **(B)** $e^{i\,5\pi/8}$ **(C)** $e^{i\,3\pi/8}$ **(D)** $e^{-i\,13\pi/8}$

Answer: (A) Detailed solution

What you need to know. $e^{2\pi i\, x/2^l}$ depends only on the fractional part of $x/2^l$, i.e. on the last l bits of x.

Step-by-step solution. $45 = 101101_2$. Dividing by 2^4 moves the binary point four places: $\frac{45}{16} = 10.1101_2 = 2 + 0.8125$. Since $e^{2\pi i\cdot 2} = 1$, only the fractional part survives:

$$e^{2\pi i\,45/16} = e^{2\pi i\,0.1101_2} = e^{2\pi i\,\frac{13}{16}} = e^{i\,13\pi/8}.$$

In QFT language: $e^{2\pi i\, x/2^l}$ depends only on the last l bits $0.x_{n-l+1}\ldots x_n$ (here $0.x_3x_4x_5x_6 = 0.1101_2$).

Why the other options are wrong.

(B) $e^{i5\pi/8}$ corresponds to $\frac{5}{16} = 0.0101_2$: it uses the wrong bits ($x_2\ldots$) instead of the last four bits 1101.

(C) $e^{i3\pi/8}$ corresponds to $\frac{3}{16}$, i.e. the bits 0011 — the bit string read wrongly.

(D) $e^{-i13\pi/8}$ has the wrong sign of the exponent; the convention uses $e^{+2\pi i\cdots}$.

Remember. Divide by 2^l = move the binary point; drop the integer part; convert the fractional part to an angle.

Q.4 SINGLE-CORRECT MCQ · Unitarity; the 4×4 Fourier matrix

The matrix $U = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & c & -1 & d \end{pmatrix}$ is unitary. Then (c, d) equals

(A) $(i,\ -i)$ **(B)** $(1,\ -1)$ **(C)** $(-1,\ 1)$ **(D)** ✓ $(-i,\ i)$

Answer: (D) Detailed solution

What you need to know. A matrix is unitary iff its rows (or columns) are orthonormal; the complex inner product conjugates the first vector.

Step-by-step solution. For a unitary matrix the rows are orthonormal. Let $r_4 = (1, c, -1, d)$.

$$\langle r_1, r_4\rangle = 1 + c - 1 + d = c + d = 0 \;\Rightarrow\; d = -c.$$

$$\langle r_2, r_4\rangle = \bar{1}\cdot 1 + \bar{i}\,c + \overline{(-1)}(-1) + \overline{(-i)}\,d = 2 - ic + id = 2 - 2ic = 0 \;\Rightarrow\; c = \tfrac{1}{i} = -i,$$

hence $d = i$. Check $\langle r_3, r_4\rangle = 1 - c - 1 - d = -(c + d) = 0$ and $\|r_4\|^2 = \frac{1}{4}(1 + 1 + 1 + 1) = 1$. U is exactly the 2-qubit QFT matrix F_4 with entries $i^{xy}/2$ (row 4: $i^{3x} = 1, -i, -1, i$).

Why the other options are wrong.

(A) $(i, -i)$ satisfies $c + d = 0$ but gives $\langle r_2, r_4\rangle = 2 - ic + id = 2 + 2 = 4 \neq 0$; it comes from forgetting the conjugate on i.

(B) $(1, -1)$ gives $\langle r_2, r_4\rangle = 2 - i - i \neq 0$.

(C) $(-1, 1)$ gives $\langle r_2, r_4\rangle = 2 + i + i \neq 0$.

Remember. Use two orthogonality conditions to solve for two unknowns, then check the norm. Recognise the result as F_4.

Q.5 SINGLE-CORRECT MCQ · Gauss sum, trace of F_4

Let F be the 4×4 unitary Fourier matrix with entries $F_{yx} = \frac{1}{2}\, i^{xy}$ $(x, y = 0, \ldots, 3)$. The trace of F is

(A) ✓ $1 + i$ **(B)** 2 **(C)** 0 **(D)** $1 - i$

Answer: (A) **Detailed solution**

What you need to know. The trace is the sum of diagonal entries; $F_{xx} = \frac{1}{2} i^{x^2}$ (a Gauss sum). Also trace = sum of eigenvalues.

Step-by-step solution. $\operatorname{tr} F = \sum_x F_{xx} = \frac{1}{2} \sum_{x=0}^{3} i^{x^2}$ (a *Gauss sum*). $x^2 = 0, 1, 4, 9$ give $i^0 = 1$, $i^1 = i$, $i^4 = 1$, $i^9 = i$. Hence $\operatorname{tr} F = \frac{1}{2}(2 + 2i) = 1 + i$. Consistency: $F^4 = I$, so its eigenvalues lie in $\{1, -1, i, -i\}$; the eigenvalues of this F are $1, 1, -1, i$ and their sum is $1 + i$.

Why the other options are wrong.

(B) 2 treats i^{x^2} as real ± 1 values; i^1 and i^9 are i, not 1.

(C) 0 confuses the trace (a sum of i^{x^2}) with the vanishing sum $\sum_x i^x$.

(D) $1 - i$ is the complex conjugate: it would be the trace of $F^\dagger = F^{-1}$.

Remember. List x^2 mod 4 and add the corresponding powers of i.

Q.6 SINGLE-CORRECT MCQ · Interference pattern $|\sum e^{ij\theta}|^2$

Let $f(\theta) = \left| 1 + e^{i\theta} + e^{2i\theta} + e^{3i\theta} \right|^2$. The maximum value of f and the smallest positive θ with $f(\theta) = 0$ are, respectively,

(A) 4 and $\pi/2$ **(B)** 4 and π **(C)** 16 and $\pi/4$ **(D)** ✓ 16 and $\pi/2$

Answer: (D) **Detailed solution**

What you need to know. $\left| \sum_{j=0}^{K-1} e^{ij\theta} \right|^2 = \frac{\sin^2(K\theta/2)}{\sin^2(\theta/2)}$: peak K^2 at $\theta = 0$, zeros at $\theta = 2\pi m/K$.

Step-by-step solution. Geometric series: $\sum_{j=0}^{3} e^{ij\theta} = \dfrac{1 - e^{4i\theta}}{1 - e^{i\theta}} = e^{3i\theta/2} \dfrac{\sin 2\theta}{\sin(\theta/2)}$, so

$$f(\theta) = \frac{\sin^2 2\theta}{\sin^2(\theta/2)}.$$

Maximum: all four arrows aligned at $\theta = 0$, $f = |4|^2 = 16$ (by the triangle inequality $f \le 16$). Zeros: $\sin 2\theta = 0$ with $\sin(\theta/2) \neq 0$, i.e. $\theta = \pi/2, \pi, 3\pi/2, \ldots$ The smallest is $\pi/2$ (four arrows pointing in the four directions $1, i, -1, -i$ cancel). This $\sin^2(K\theta/2)/\sin^2(\theta/2)$ shape is the peak shape of every QFT output.

Why the other options are wrong.

(A) 4 and $\pi/2$: 4 is the maximum of $|\sum|$ (the modulus), not of $|\sum|^2$.

(B) 4 and π: wrong maximum, and π is a zero but not the smallest positive one.

(C) 16 and $\pi/4$: at $\theta = \pi/4$ the four arrows span only 135° and do not cancel ($f(\pi/4) = \sin^2(\pi/2)/\sin^2(\pi/8) \approx 6.83 \neq 0$; the zeros are at multiples of $2\pi/4$).

Remember. K arrows cancel first when they are spread evenly round the circle: $\theta = 2\pi/K$.

Q.7 SINGLE-CORRECT MCQ · Continued fractions, convergents

The best rational approximation p/q to $\dfrac{11}{64}$ with $1 \le q \le 10$ (as produced by the continued-fraction algorithm) is

(A) 2/9 **(B)** ✓ 1/6 **(C)** 1/7 **(D)** 1/5

Answer: (B) Detailed solution

What you need to know. Continued fraction expansion and its convergents give the best rational approximations with bounded denominator.

Step-by-step solution. Continued fraction of $\frac{11}{64}$: $64 = 5 \cdot 11 + 9$, $11 = 1 \cdot 9 + 2$, $9 = 4 \cdot 2 + 1$, $2 = 2 \cdot 1$, so $\frac{11}{64} = [0; 5, 1, 4, 2]$. Convergents: $\frac{0}{1}, \frac{1}{5}, \frac{1}{6}, \frac{5}{29}, \frac{11}{64}$. The last convergent with $q \le 10$ is $\frac{1}{6}$. Direct check: $\left|\frac{11}{64} - \frac{1}{6}\right| = \frac{2}{384} \approx 0.0052$, while $\left|\frac{11}{64} - \frac{1}{5}\right| \approx 0.028$, $\frac{1}{7}$ differs by 0.029 and $\frac{2}{9}$ by 0.050. This is exactly the step that turns a QFT measurement $y = 11$ out of $M = 64$ into the period $r = 6$.

Why the other options are wrong.

(A) 2/9 is not a convergent and is far away (0.050).

(C) 1/7 is not a convergent; its error 0.029 is much larger than that of 1/6.

(D) 1/5 is an earlier convergent; 1/6 has an allowed denominator and is closer (0.0052 vs 0.028).

Remember. Compute $[a_0; a_1, \ldots]$, build convergents, keep the last one with denominator below the bound.

Q.8 SINGLE-CORRECT MCQ · Multiplicative order; Shor's gcd step

The order r of $a = 3$ modulo $N = 35$ (the least $r > 0$ with $a^r \equiv 1 \pmod{35}$), and the numbers $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$, are

(A) $r = 6$; 5 and 7 **(B)** $r = 12$; 1 and 35

(C) ✓ $r = 12$; 7 and 5 **(D)** $r = 4$; 5 and 7

Answer: (C) Detailed solution

What you need to know. Order r: least power with $a^r \equiv 1$; $r \mid \varphi(N)$. Shor's step: if r even and $a^{r/2} \not\equiv -1$, $\gcd(a^{r/2} \pm 1, N)$ are factors.

Step-by-step solution. Powers of 3 modulo 35: $3, 9, 27, 11, 33, 29, 17, 16, 13, 4, 12, 1$, so $r = 12$ (by Euler, $r \mid \varphi(35) = 24$; $3^6 \equiv 29 \not\equiv 1$ and $3^4 \equiv 11 \not\equiv 1$, so r is neither 6 nor 4). $a^{r/2} = 3^6 \equiv 29 \pmod{35}$, which is not $\equiv -1 = 34$, so Shor's reduction works:

$$\gcd(29 - 1, 35) = \gcd(28, 35) = 7, \qquad \gcd(29 + 1, 35) = \gcd(30, 35) = 5.$$

Indeed $35 = 5 \times 7$.

Why the other options are wrong.

(A) $r = 6$ is wrong because $3^6 \equiv 29 \not\equiv 1 \pmod{35}$.

(B) $r = 12$ with gcds 1 and 35 would mean $a^{r/2} \equiv \pm 1$; but $3^6 \equiv 29$, and the gcds are 7 and 5.

(D) $r = 4$ is wrong because $3^4 = 81 \equiv 11 \not\equiv 1$.

Remember. List powers until you return to 1; test candidates that divide $\varphi(N)$; then take the two gcds.

Q.9 SINGLE-CORRECT MCQ · Parameters of an n-qubit state

A general pure state of 10 qubits is described by D complex amplitudes. After removing normalisation and the unobservable global phase, the number of independent *real* parameters is P. Then (D, P) is

(A) $(1024,\ 2048)$ **(B)** ✓ $(1024,\ 2046)$ **(C)** $(1024,\ 2047)$ **(D)** $(20,\ 40)$

Answer: (B) **Detailed solution**

What you need to know. n qubits $\Rightarrow 2^n$ complex amplitudes; normalisation and global phase each remove one real parameter.

Step-by-step solution. n qubits span a 2^n-dimensional space: $D = 2^{10} = 1024$ complex numbers $= 2048$ real numbers. Normalisation $\sum |\alpha_x|^2 = 1$ removes one real parameter and the global phase $e^{i\gamma}$ removes another: $P = 2048 - 2 = 2046$. (Option $(1024, 2047)$ forgets the global phase; $(20, 40)$ counts as if qubits were independent.)

Why the other options are wrong.

(A) $(1024, 2048)$ removes neither normalisation nor global phase.

(C) $(1024, 2047)$ removes normalisation but forgets that the global phase is unobservable.

(D) $(20, 40)$ counts as if the qubits were independent (product states only); it ignores entanglement.

Remember. $2^{n+1} - 2$ real parameters: exponential in n, the root of quantum computing's power.

Q.10 SINGLE-CORRECT MCQ · Walsh–Hadamard signs

For $n = 4$, $H^{\otimes 4}|x\rangle = \frac{1}{4}\sum_y (-1)^{x\cdot y}|y\rangle$ where $x \cdot y = \sum_j x_j y_j$. For $x = 1011_2$, the number of $y \in \{0, \ldots, 15\}$ with a *negative* amplitude is

(A) 4 **(B)** ✓ 8 **(C)** 12 **(D)** 3

Answer: (B) **Detailed solution**

What you need to know. $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{N}}\sum_y (-1)^{x\cdot y}|y\rangle$; the sign is negative iff $x \cdot y$ is odd.

Step-by-step solution. The amplitude is negative iff $x \cdot y = y_1 + y_3 + y_4$ is odd (because $x = 1011$ has $x_1 = x_3 = x_4 = 1$). y_2 is free (2 choices) and among the 8 choices of (y_1, y_3, y_4) exactly half have odd sum, so the count is $2 \times 4 = 8$. In general, for any $x \neq 0$ exactly 2^{n-1} of the 2^n signs are negative (the characters $(-1)^{x\cdot y}$ of $\mathbb{Z}_2^n$ sum to zero).

Why the other options are wrong.

(A) 4 ignores the free bit y_2 (which doubles the count).

(C) 12 counts the wrong parity or adds the free bit twice.

(D) 3 is just the number of 1-bits in x, unrelated to the sign count.

Remember. For any $x \neq 0$, exactly half of the 2^n Hadamard signs are negative.

Q.11 Single-correct MCQ · FFT vs. QFT cost (and its limits)

A classical FFT on a vector of length 2^n needs about $n\,2^n$ operations, while the QFT circuit needs $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor$ gates. For $n = 20$ the ratio (FFT operations)/(QFT gates) is closest to

(A) 10^9 **(B)** 10^3 **(C)** ✓ 10^5 **(D)** 10^7

Answer: (C) **Detailed solution**

What you need to know. FFT cost $n2^n$; QFT gates $\frac{n(n+1)}{2}+\lfloor n/2 \rfloor$. Compare orders of magnitude.
Step-by-step solution. FFT: $20 \times 2^{20} = 20 \times 1\,048\,576 = 20\,971\,520$. QFT: $\frac{20\cdot 21}{2} + 10 = 210 + 10 = 220$ gates. Ratio $= 20\,971\,520/220 \approx 9.5 \times 10^4 \approx 10^5$. Caveat (important for interpretation): the QFT output is stored in amplitudes and cannot be read out completely, so this ratio is *not* a speed-up for computing all coefficients.
Why the other options are wrong.
(A) 10^9 uses N^2 or similar instead of $n2^n$.
(B) 10^3 underestimates: $2.1 \times 10^7/220 \approx 9.5 \times 10^4$.
(D) 10^7 forgets to divide by the QFT gate count.
Remember. Exponentially fewer gates — but the QFT output cannot be read out completely, so this is not a speed-up for computing all coefficients.

Q.12 Single-correct MCQ · Cyclotomic product identity

With $\omega = e^{2\pi i/8}$, the value of $\prod_{k=1}^{7} |1 - \omega^k|$ is

(A) ✓ 8 **(B)** 0 **(C)** 1 **(D)** 7

Answer: (A) **Detailed solution**

What you need to know. $z^N - 1 = \prod_k (z - \omega^k)$, so $1 + z + \cdots + z^{N-1} = \prod_{k \geq 1}(z - \omega^k)$; set $z = 1$.
Step-by-step solution. The roots of $z^8 - 1$ are $\omega^0, \ldots, \omega^7$, so $z^8 - 1 = \prod_{k=0}^{7}(z - \omega^k)$ and

$$\frac{z^8 - 1}{z - 1} = 1 + z + \cdots + z^7 = \prod_{k=1}^{7}(z - \omega^k).$$

Put $z = 1$: $\prod_{k=1}^{7}(1 - \omega^k) = 8$, hence the product of the moduli is 8. (Geometrically: the product of the distances from one vertex of a regular octagon inscribed in the unit circle to the other seven vertices.)
Why the other options are wrong.
(B) 0 would require some $\omega^k = 1$ for $k = 1, \ldots, 7$, which is false.
(C) 1 wrongly assumes the product of distances is 1 because each $|\omega^k| = 1$.
(D) 7 counts the factors instead of evaluating the polynomial at $z = 1$.
Remember. Products of $(1 - \omega^k)$ come from the factorisation of $z^N - 1$.

Q.13 SINGLE-CORRECT MCQ · DFT of a vector; Parseval

The normalised vector $\alpha = \frac{1}{3}(1,2,2,0)$ is transformed by the unitary 4×4 Fourier matrix $\beta_y = \frac{1}{2}\sum_x \alpha_x\, i^{xy}$. The value of $|\beta_0|^2$ is

(A) $\frac{5}{6}$ **(B)** $\frac{25}{12}$ **(C)** $\frac{1}{4}$ **(D)** ✓ $\frac{25}{36}$

Answer: (D) Detailed solution

What you need to know. $\beta_0 = \frac{1}{\sqrt{N}}\sum_x \alpha_x$; probability = |amplitude|^2; Parseval $\sum|\beta|^2 = 1$.

Step-by-step solution. $\beta_0 = \frac{1}{2}\sum_x \alpha_x = \frac{1}{2}\cdot\frac{1+2+2+0}{3} = \frac{5}{6}$, so $|\beta_0|^2 = \frac{25}{36}$. Check the other options: $\frac{5}{6}$ is the amplitude, not the probability; $\frac{25}{12} > 1$ is impossible because $\sum_y |\beta_y|^2 = \sum_x |\alpha_x|^2 = 1$ (Parseval, F is unitary). The remaining probability $\frac{11}{36}$ is spread over $y = 1,2,3$.

Why the other options are wrong.

(A) $\frac{5}{6}$ is the amplitude β_0, not its squared modulus.

(B) $\frac{25}{12} > 1$ cannot be a probability; it forgets the factor $\frac{1}{2}$.

(C) $\frac{1}{4}$ assumes a uniform output, true only for basis-state inputs.

Remember. Zero frequency = average of the amplitudes; always square at the end.

Q.14 SINGLE-CORRECT MCQ · Binary expansion of $2^k\varphi$

Let $\varphi = 0.\varphi_1\varphi_2\varphi_3$ (binary, $\varphi_j \in \{0,1\}$). Then $e^{2\pi i\, 4\varphi}$ equals

(A) $e^{2\pi i\, 0.\varphi_2\varphi_3}$ **(B)** i^{φ_2} **(C)** $(-1)^{\varphi_1}$ **(D)** ✓ $(-1)^{\varphi_3}$

Answer: (D) Detailed solution

What you need to know. Multiplying a binary fraction by 2^k moves the binary point k places right; whole turns vanish.

Step-by-step solution. $4\varphi = \varphi_1\varphi_2.\varphi_3$ in binary, i.e. $4\varphi = (2\varphi_1+\varphi_2) + 0.\varphi_3$. The integer part is a whole number of turns, so $e^{2\pi i\, 4\varphi} = e^{2\pi i\, 0.\varphi_3} = e^{i\pi\varphi_3} = (-1)^{\varphi_3}$. This is why, in phase estimation, the counting qubit controlling U^4 carries only the *last* bit φ_3, and the inverse QFT reads the bits from there.

Why the other options are wrong.

(A) $e^{2\pi i\, 0.\varphi_2\varphi_3}$ is $e^{2\pi i\, 2\varphi}$ (shift by one place), not by two.

(B) i^{φ_2} mixes the wrong bit and the wrong angle.

(C) $(-1)^{\varphi_1}$ would be $e^{2\pi i\varphi}$ for $\varphi = 0.\varphi_1$ only; after multiplying by 4 the bit φ_1 is in the integer part.

Remember. U^{2^k} in phase estimation sees only the bits after position k.

Q.15 MATCHING LISTS · Algebra of 8th roots of unity

Let $\omega = e^{2\pi i/8}$. Match List-I with List-II.

List-I		List-II	
(P)	ω^4	**(1)**	$\sqrt{2}$
(Q)	$\sum_{k=0}^{7}\omega^{2k}$	**(2)**	-1
(R)	$\omega\cdot\omega^7$	**(3)**	0
(S)	$\omega+\omega^7$	**(4)**	1

(A) P→1, Q→3, R→4, S→2 **(B)** P→2, Q→4, R→3, S→1

(C) ✓ P→2, Q→3, R→4, S→1 **(D)** P→3, Q→2, R→1, S→4

Answer: (C) **Detailed solution**

What you need to know. Powers of $\omega = e^{i\pi/4}$: $\omega^2 = i$, $\omega^4 = -1$, $\omega^8 = 1$, $\omega + \bar{\omega} = 2\cos(\pi/4)$.
Step-by-step solution. P: $\omega^4 = e^{i\pi} = -1$ (2). Q: $\omega^2 = i$ is a primitive 4th root, and $\sum_{k=0}^{7} i^k = 2(1 + i - 1 - i) = 0$ (3). R: $\omega\,\omega^7 = \omega^8 = 1$ (4). S: $\omega + \omega^7 = \omega + \bar{\omega} = 2\cos\frac{\pi}{4} = \sqrt{2}$ (1). Hence P→2, Q→3, R→4, S→1.
Why the other options are wrong.
(A) P→1 says $\omega^4 = \sqrt{2}$; it confuses a power with the sum $\omega + \omega^7$.
(B) Q→4 says $\sum \omega^{2k} = 1$; the sum of a full cycle of powers of i is 0.
(D) P→3 says $\omega^4 = 0$; a root of unity has modulus 1, never 0.
Remember. In matching questions solve the easiest row (here R: $\omega^8 = 1$) and eliminate options.

Q.16 MATCHING LISTS · Properties of F, F^2, $H \otimes H$, CT

F is the 4×4 unitary Fourier matrix ($F_{yx} = i^{xy}/2$), H the Hadamard matrix and $CT = \mathrm{diag}(1, 1, 1, e^{i\pi/4})$. Match each matrix in List-I with the *only* property in List-II that it has.

List-I		List-II	
(P)	F	**(1)**	a permutation matrix that is its own inverse
(Q)	F^2	**(2)**	real, symmetric, its own inverse, but not a permutation matrix
(R)	$H \otimes H$	**(3)**	diagonal with determinant $e^{i\pi/4}$
(S)	CT	**(4)**	$M^4 = I$ but $M^2 \neq I$

(A) P→4, Q→2, R→1, S→3
(B) ✓ P→4, Q→1, R→2, S→3
(C) P→3, Q→1, R→2, S→4
(D) P→1, Q→4, R→2, S→3

Answer: (B) **Detailed solution**

What you need to know. $F^2\,|x\rangle = |-x\rangle$, $F^4 = I$; $H \otimes H$ is real symmetric self-inverse; det of a diagonal = product of entries.
Step-by-step solution. $(F^2)_{zx} = \frac{1}{4}\sum_y i^{y(x+z)} = \delta_{z,\,-x \bmod 4}$, so F^2 is the permutation $|x\rangle \mapsto |-x\rangle$ ($0 \to 0$, $1 \leftrightarrow 3$, $2 \to 2$), which squares to I: Q→1. Hence $F^4 = I$ while $F^2 \neq I$: P→4. $H \otimes H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ is real, symmetric and $(H \otimes H)^2 = I$, but its entries are $\pm\frac{1}{2}$, so it is not a permutation: R→2. CT is diagonal with det $= e^{i\pi/4}$; note $CT^4 = \mathrm{diag}(1, 1, 1, -1) \neq I$, so it does not have property 4: S→3.
Why the other options are wrong.
(A) Q→2 says F^2 is not a permutation; but F^2 maps $|x\rangle \mapsto |-x\rangle$, a permutation.
(C) P→3 says F is diagonal; F is a full matrix.
(D) P→1 says F is a permutation; F has entries of modulus $\frac{1}{2}$, not 0/1.
Remember. Squaring the Fourier matrix gives the reflection $x \mapsto -x$.

Q.17 Assertion–Reason · Root sums when $\gcd(k, N) > 1$

Assertion (A): For every integer k that is not a multiple of N, $\sum_{j=0}^{N-1} e^{2\pi i\, jk/N} = 0$.

Reason (R): For every integer k that is not a multiple of N, the N numbers $e^{2\pi i\, jk/N}$ $(j = 0, \ldots, N-1)$ are the N distinct vertices of a regular N-gon centred at the origin.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) ✓ A is true, but R is false

(D) A is false, but R is true

Answer: (C) Detailed solution

What you need to know. Assertion-Reason: decide A and R separately, then whether R explains A. Roots repeat when $\gcd(k, N) > 1$.

Step-by-step solution. **A is true:** with $z = e^{2\pi ik/N} \neq 1$, $\sum_{j=0}^{N-1} z^j = \frac{1-z^N}{1-z} = \frac{1-e^{2\pi ik}}{1-z} = 0$. **R is false:** if $g = \gcd(k, N) > 1$ the numbers repeat. Example $N = 8$, $k = 2$: the values are $1, i, -1, -i, 1, i, -1, -i$ — a square traversed twice, not 8 distinct vertices of an octagon. (The sum is still zero, because a regular N/g-gon is traversed g times.) Answer: (C).

Why the other options are wrong.

(A) needs A true, R true *and* R explaining A; R is false (counter-example $N = 8$, $k = 2$).

(B) needs A and R both true but R not explaining A; R is false.

(D) needs A false and R true; A is true (geometric series).

Remember. A true statement can be explained by a false-looking "intuitive" reason; check the reason on an example.

Q.18 Statement I / II · Determinants of unitary matrices

Statement I: Every unitary matrix U satisfies $|\det U| = 1$.

Statement II: The unitary Fourier matrix F_N $(N = 2^n$, entries $\omega^{xy}/\sqrt{N})$ has $\det F_N = \pm 1$ for every n.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

What you need to know. $|\det U| = 1$ for unitaries; $F^4 = I$ so the eigenvalues lie in $\{\pm 1, \pm i\}$.

Step-by-step solution. **I is true:** $|\det U|^2 = \det(U^\dagger U) = \det I = 1$. **II is false:** the determinant is the product of the eigenvalues, which lie in $\{1, -1, i, -i\}$ because $F^4 = I$. For $N = 4$ the eigenvalues are $1, 1, -1, i$ (compare the trace $1 + i$ in Q.5), so $\det F_4 = 1 \cdot 1 \cdot (-1) \cdot i = -i \neq \pm 1$. (For $N = 2$, $F_2 = H$ and $\det H = -1$, so the claim happens to hold there — one counterexample is enough.) Answer: (C).

Why the other options are wrong.

(A) needs both statements true; II fails for $N = 4$ ($\det F_4 = -i$).

(B) needs both statements false; I is true.

(D) needs I false and II true; I is true and II is false.

Remember. Modulus 1 does not mean real: a determinant can be $\pm i$.

Paragraph for Questions 19 and 20

Let $N = 32$ and define the sequence $a_x = 1$ if $x \equiv 2 \pmod 5$ and $a_x = 0$ otherwise, for $x = 0, 1, \ldots, 31$. Its (un-normalised) Fourier coefficients are $A_y = \sum_{x=0}^{31} a_x\, e^{2\pi i\, xy/32}$, $y = 0, 1, \ldots, 31$. (The sequence is "periodic with period 5", but 5 does not divide 32.)

Q.19 PARAGRAPH BASED · Parseval for the un-normalised DFT

The value of $\displaystyle\sum_{y=0}^{31} |A_y|^2$ is

(A) ✓ 192 **(B)** 6 **(C)** 36 **(D)** 1152

Answer: (A) Detailed solution

What you need to know. Parseval for the un-normalised transform: $\sum_y |A_y|^2 = N \sum_x |a_x|^2$.
Step-by-step solution. The non-zero samples are $x = 2, 7, 12, 17, 22, 27$, i.e. $K = 6$ ones. For the un-normalised transform $A = \sqrt{N}\, Fa$ with unitary F, Parseval gives

$$\sum_y |A_y|^2 = N \sum_x |a_x|^2 = 32 \times 6 = 192.$$

(36 is $|A_0|^2$ alone; 6 is $\sum_x |a_x|^2$; 1152 would be $N \cdot 36$.)
Why the other options are wrong.
(B) 6 is $\sum_x |a_x|^2$ without the factor N.
(C) 36 is only $|A_0|^2$.
(D) $1152 = 32 \times 36$ multiplies the wrong quantity.
Remember. Count the non-zero samples, multiply by N.

Q.20 PARAGRAPH BASED · Spectral leakage ($r \nmid N$)

Among $y = 1, 2, \ldots, 15$, the value of y for which $|A_y|$ is largest is

(A) 6
(B) 16 is the only peak; all of $1, \ldots, 15$ give $|A_y| = 0$
(C) 7
(D) ✓ 13

Answer: (D) Detailed solution

What you need to know. With period $r \nmid N$ the peaks leak; $|A_y| = |\sin(\pi K r y/N)/\sin(\pi r y/N)|$ is largest when ry/N is closest to an integer.
Step-by-step solution. $A_y = \sum_{j=0}^{5} e^{2\pi i(5j+2)y/32} = e^{4\pi i y/32} \sum_{j=0}^{5} e^{2\pi i\, 5jy/32}$, so

$$|A_y| = \left|\frac{\sin(6 \cdot 5\pi y/32)}{\sin(5\pi y/32)}\right| = \left|\frac{\sin(30\pi y/32)}{\sin(5\pi y/32)}\right|.$$

Peaks occur where $5y/32$ is close to an integer, i.e. $y \approx 32k/5 = 6.4,\ 12.8,\ \ldots$ The closer $5y/32$ is to an integer, the higher the peak: $5 \cdot 13/32 = 2.03$ (distance 0.03) beats $5 \cdot 6/32 = 0.94$ (distance 0.06) and $5 \cdot 7/32 = 1.09$. Numerically $|A_{13}| = 5.67$, $|A_6| = 4.74$, $|A_7| = 3.38$, $|A_{12}| = 1.85$; and $|A_{16}| = 0$ since $\sin(15\pi) = 0$. Hence $y = 13$. Lesson: with a period that does not divide N, the QFT peaks "leak" and their heights depend on how close ky/N is to an integer.
Why the other options are wrong.
(A) $y = 6$: $5 \cdot 6/32 = 0.94$ is 0.06 from an integer, twice as far as $y = 13$ (0.03); $|A_6| = 4.74 < 5.67$.
(B) $|A_y|$ is not zero for $y = 1..15$; and $y = 16$ actually gives $|A_{16}| = 0$ ($\sin 15\pi = 0$).

(C) $y = 7$: $5 \cdot 7/32 = 1.09$ is 0.09 away; $|A_7| = 3.38$.
Remember. Rank peaks by the distance of ry/N from the nearest integer.

SILICO FELLER

SECTION B · Quantum Mechanics Q.21–40

Q.21 SINGLE-CORRECT MCQ · Born rule, partial measurement

A two-qubit state is $|\psi\rangle = c\,(\,|00\rangle + 2i\,|01\rangle - |10\rangle + \sqrt{2}\,|11\rangle\,)$ with $c > 0$. If only the *first* qubit is measured in the computational basis, the probability of obtaining 1 is

(A) $\frac{1}{2}$ **(B)** ✓ $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

Answer: (B) Detailed solution

What you need to know. Normalise first; partial measurement probability = sum of $|c_x|^2$ over basis states with the required bit.

Step-by-step solution. Normalisation: $c^2(|1|^2+|2i|^2+|-1|^2+|\sqrt{2}|^2) = c^2(1+4+1+2) = 8c^2 = 1$, so $c = \frac{1}{2\sqrt{2}}$. By the Born rule, the probability that the first qubit gives 1 is the total weight of the basis states whose first bit is 1:

$$P(1) = c^2(|-1|^2 + |\sqrt{2}|^2) = \frac{1+2}{8} = \frac{3}{8}.$$

(The phases i and -1 do not affect probabilities in this basis; $\frac{5}{8}$ is $P(0)$; $\frac{1}{4}$ and $\frac{1}{2}$ come from forgetting to normalise or from counting states instead of weights.)

Why the other options are wrong.

(A) $\frac{1}{2}$ counts basis states (2 of 4) instead of weights.

(C) $\frac{1}{4}$ forgets to normalise or uses only one of the two terms.

(D) $\frac{5}{8}$ is $P(\text{first qubit} = 0)$.

Remember. Probabilities come from squared moduli of normalised amplitudes; phases do not matter here.

Q.22 SINGLE-CORRECT MCQ · Global vs. relative phase

Which pair of single-qubit state vectors describes the *same* physical state (i.e. gives identical measurement statistics in every basis)?

(A) ✓ $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle - |1\rangle$

(B) $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle + |1\rangle$

(C) $|0\rangle + i\,|1\rangle$ and $|0\rangle - i\,|1\rangle$

(D) $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$

Answer: (A) Detailed solution

What you need to know. Two vectors describe the same state iff one is a constant phase times the other (component ratios all equal).

Step-by-step solution. Two vectors represent the same physical state iff they differ by an overall (global) factor $e^{i\gamma}$. $i\,|0\rangle - |1\rangle = i(\,|0\rangle + i\,|1\rangle\,)$, since $i \cdot i = -1$: global phase i, so the pair is physically identical. The other pairs differ by a *relative* phase: $|0\rangle \pm i\,|1\rangle$ (relative phase $\pm i$), $|0\rangle \pm |1\rangle$ (relative phase ± 1), and $i\,|0\rangle + |1\rangle = i(|0\rangle - i\,|1\rangle)$, which is $|0\rangle - i\,|1\rangle$ again. A relative phase is measurable: e.g. $|0\rangle + |1\rangle \propto |+\rangle$ and $|0\rangle - |1\rangle \propto |-\rangle$ are perfectly distinguishable in the X basis. Relative phases are exactly what the QFT writes information into.

Why the other options are wrong.

(B) $i\,|0\rangle + |1\rangle = i(|0\rangle - i\,|1\rangle)$, which equals $|0\rangle - i\,|1\rangle$, not $|0\rangle + i\,|1\rangle$.

(C) $|0\rangle \pm i\,|1\rangle$: ratios 1 and -1 differ (relative phase) — they are orthogonal states.

(D) $|0\rangle \pm |1\rangle$ are $|+\rangle$ and $|-\rangle$, orthogonal.

Remember. Global phase: invisible. Relative phase: physical.

Q.23 Single-correct MCQ · Interference with a Hadamard

The state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$ passes through a Hadamard gate and is then measured in the computational basis. The probability of outcome 0 is

(A) ✓ $\cos^2(\phi/2)$ **(B)** $\sin^2(\phi/2)$ **(C)** $\frac{1}{2}$ **(D)** $\cos^2\phi$

Answer: (A) Detailed solution

What you need to know. H maps relative phase to probability: amplitudes $\frac{1\pm e^{i\phi}}{2}$; $|1+e^{i\phi}|^2 = 4\cos^2(\phi/2)$.

Step-by-step solution. With $H|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $H|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$:

$$H\,\frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}} = \frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle.$$

$|1+e^{i\phi}|^2 = (1+e^{i\phi})(1+e^{-i\phi}) = 2+2\cos\phi = 4\cos^2(\phi/2)$, so $P(0) = \frac{4\cos^2(\phi/2)}{4} = \cos^2(\phi/2)$ (and $P(1) = \sin^2(\phi/2)$). The Hadamard converts an invisible *relative phase* into a visible *probability*: $\phi = 0$ gives 0 surely, $\phi = \pi$ gives 1 surely. A single-qubit QFT is exactly this H.

Why the other options are wrong.

(B) $\sin^2(\phi/2)$ is $P(1)$.

(C) $\frac{1}{2}$ ignores the interference produced by H (true only for $\phi = \pm\pi/2$).

(D) $\cos^2\phi$ drops the half-angle.

Remember. Interference converts phase into measurable probability.

Q.24 Single-correct MCQ · Bloch-sphere expectation values

For the qubit state $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$ the quantity $\langle X\rangle + i\langle Y\rangle$, where $\langle A\rangle = \langle\psi|A|\psi\rangle$ and X, Y are Pauli matrices, equals

(A) $\cos\theta\, e^{i\phi}$ **(B)** ✓ $\sin\theta\, e^{i\phi}$ **(C)** $\sin\frac{\theta}{2}\, e^{i\phi}$ **(D)** $\sin\theta\, e^{-i\phi}$

Answer: (B) Detailed solution

What you need to know. $\langle X\rangle = 2\mathrm{Re}(\bar{a}b)$, $\langle Y\rangle = 2\mathrm{Im}(\bar{a}b)$, so $\langle X\rangle + i\langle Y\rangle = 2\bar{a}b$.

Step-by-step solution. Write $a = \cos\frac{\theta}{2}$, $b = e^{i\phi}\sin\frac{\theta}{2}$. With $X = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, $Y = \left(\begin{smallmatrix}0&-i\\i&0\end{smallmatrix}\right)$:

$$\langle X\rangle = \bar{a}b + \bar{b}a = 2\,\mathrm{Re}(\bar{a}b), \qquad \langle Y\rangle = -i\bar{a}b + i\bar{b}a = 2\,\mathrm{Im}(\bar{a}b).$$

Hence $\langle X\rangle + i\langle Y\rangle = 2\bar{a}b = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\,e^{i\phi} = \sin\theta\, e^{i\phi}$. So $\langle X\rangle = \sin\theta\cos\phi$, $\langle Y\rangle = \sin\theta\sin\phi$ (and $\langle Z\rangle = \cos\theta$): the Bloch vector $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. The *azimuth* ϕ is the phase that QFT rotations change.

Why the other options are wrong.

(A) $\cos\theta\, e^{i\phi}$ uses $\langle Z\rangle$'s factor instead of $2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$.

(C) $\sin\frac{\theta}{2}\, e^{i\phi}$ forgets the factor $2\cos\frac{\theta}{2}$.

(D) $\sin\theta\, e^{-i\phi}$ is $\langle X\rangle - i\langle Y\rangle$ (conjugate).

Remember. Bloch vector $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$.

Q.25 SINGLE-CORRECT MCQ · Entanglement test

Which of the following two-qubit states is *entangled* (cannot be written as $|a\rangle \otimes |b\rangle$)?

(A) $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
(B) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
(C) ✓ $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$
(D) $\frac{1}{2}(|00\rangle + i|01\rangle - |10\rangle - i|11\rangle)$

Answer: (C) Detailed solution

What you need to know. Product test: $c_{00}c_{11} = c_{01}c_{10}$.
Step-by-step solution. For $\sum c_{jk}|jk\rangle$, a product $(a_0|0\rangle + a_1|1\rangle)(b_0|0\rangle + b_1|1\rangle)$ has $c_{jk} = a_j b_k$, so $c_{00}c_{11} = a_0a_1b_0b_1 = c_{01}c_{10}$. Conversely, $c_{00}c_{11} = c_{01}c_{10}$ (the 2×2 coefficient matrix has determinant 0, i.e. rank 1) guarantees a product. Test each option (factor $\frac{1}{2}$ dropped):

- $(1,1,1,-1)$: $c_{00}c_{11} = -1 \neq 1 = c_{01}c_{10} \Rightarrow$ **entangled**.
- $(1,1,1,1)$: $1 = 1$, product $|+\rangle|+\rangle$.
- $(1,-1,1,-1)$: $-1 = -1$, product $|+\rangle|-\rangle$.
- $(1,i,-1,-i)$: $-i = -i$, product $\frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + i|1\rangle)$.

The entangled state is $\mathrm{CZ}|+\rangle|+\rangle$: a controlled-phase gate (like those in the QFT) *can* entangle. (This is not in conflict with Q.54: the QFT acting on a *basis* state happens to give a product state.)
Why the other options are wrong.
(A) $(1,-1,1,-1)/2 = |+\rangle|-\rangle$: $-1 = -1$, product.
(B) $(1,1,1,1)/2 = |+\rangle|+\rangle$: $1 = 1$, product.
(D) $(1,i,-1,-i)/2 = \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + i|1\rangle)$: $-i = -i$, product.
Remember. One sign change in the uniform state creates entanglement ($CZ|+\rangle|+\rangle$).

Q.26 SINGLE-CORRECT MCQ · Phase kickback (CNOT)

A CNOT gate (qubit 1 = control, qubit 2 = target) acts on $|+\rangle|-\rangle$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. The output is

(A) $|+\rangle|-\rangle$
(B) $|-\rangle|+\rangle$
(C) $\frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|+\rangle)$
(D) ✓ $|-\rangle|-\rangle$

Answer: (D) Detailed solution

What you need to know. $|-\rangle$ is an eigenvector of X with eigenvalue -1; controlled gates kick eigenvalues back to the control.
Step-by-step solution. $|-\rangle$ is an eigenvector of X with eigenvalue -1: $X|-\rangle = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$. Therefore

$$\mathrm{CNOT}\,|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle\, X|-\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|-\rangle = |-\rangle|-\rangle.$$

The target is unchanged; the eigenvalue -1 has been "kicked back" onto the control as a relative phase. This *phase kickback* is the mechanism by which controlled-U^{2^j} gates load the eigenphase into the counting register in phase estimation (see Q.28).
Why the other options are wrong.
(A) $|+\rangle|-\rangle$ assumes CNOT does nothing because the target is unchanged — but the control acquires a phase.
(B) $|-\rangle|+\rangle$ flips the wrong qubit.
(C) $\frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|+\rangle)$ treats $X|-\rangle = |+\rangle$, which is false ($X|-\rangle = -|-\rangle$).
Remember. Phase kickback: the target stays, the control changes.

Q.27 SINGLE-CORRECT MCQ · Controlled-phase matrices

In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (first qubit = control), the matrix of controlled-R_3, where $R_k = \mathrm{diag}\,(1, e^{2\pi i/2^k})$, is

(A) ✓ $\mathrm{diag}\,(1,1,1,e^{i\pi/4})$ **(B)** $\mathrm{diag}\,(1,e^{i\pi/4},1,e^{i\pi/4})$

(C) $\mathrm{diag}\,(1,1,e^{i\pi/4},e^{i\pi/4})$ **(D)** $\mathrm{diag}\,(1,1,1,e^{i\pi/8})$

Answer: (A) Detailed solution

What you need to know. $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$; $R_3 = \mathrm{diag}(1, e^{2\pi i/8})$.
Step-by-step solution. $R_3 = \mathrm{diag}(1, e^{2\pi i/8}) = \mathrm{diag}(1, e^{i\pi/4})$ (the T gate). Controlled-R_3 applies R_3 to the target only when the control is $|1\rangle$:

$$CR_3 = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_3 = \mathrm{diag}(1,1,1,e^{i\pi/4}).$$

Only $|11\rangle$ picks up a phase, so the gate is *symmetric*: it does not matter which qubit is called the control. $\mathrm{diag}(1,e^{i\pi/4},1,e^{i\pi/4}) = I \otimes R_3$ and $\mathrm{diag}(1,1,e^{i\pi/4},e^{i\pi/4}) = R_3 \otimes I$ are uncontrolled; $e^{i\pi/8}$ confuses $2\pi/2^3$ with $\pi/2^3$.
Why the other options are wrong.
(B) $\mathrm{diag}(1,e^{i\pi/4},1,e^{i\pi/4}) = I \otimes R_3$ (uncontrolled).
(C) $\mathrm{diag}(1,1,e^{i\pi/4},e^{i\pi/4}) = R_3 \otimes I$ (uncontrolled).
(D) $e^{i\pi/8}$ uses $\pi/2^3$ instead of $2\pi/2^3$.
Remember. A controlled phase changes only $|11\rangle$, and is symmetric in the two qubits.

Q.28 SINGLE-CORRECT MCQ · Kickback of controlled-U^{2^j}

$U|u\rangle = e^{2\pi i\varphi}|u\rangle$. A control qubit in $|+\rangle$ and the register in $|u\rangle$ pass through a controlled-U^{2^j} gate. The output state is

(A) $\frac{1}{\sqrt{2}} e^{2\pi i\, 2^j\varphi}(\,|0\rangle + |1\rangle\,)\,|u\rangle$ **(B)** ✓ $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, 2^j\varphi}\,|1\rangle\,)\,|u\rangle$

(C) $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, j\varphi}\,|1\rangle\,)\,|u\rangle$ **(D)** $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,\varphi/2^j}\,|1\rangle\,)\,|u\rangle$

Answer: (B) Detailed solution

What you need to know. $U^m|u\rangle = e^{2\pi i m\varphi}|u\rangle$; the controlled gate acts only on the $|1\rangle$ branch.
Step-by-step solution. $U^{2^j}|u\rangle = (e^{2\pi i\varphi})^{2^j}|u\rangle = e^{2\pi i\, 2^j\varphi}|u\rangle$. The gate acts only on the $|1\rangle$ branch:

$$\frac{1}{\sqrt{2}}(\,|0\rangle\,|u\rangle + |1\rangle\; U^{2^j}\,|u\rangle\,) = \frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, 2^j\varphi}\,|1\rangle\,)\,|u\rangle\,.$$

The phase is *relative* (only on $|1\rangle$), not global, so it is measurable after a Hadamard / inverse QFT. If $\varphi = 0.\varphi_1\varphi_2\ldots$ in binary, $2^j\varphi$ shifts the binary point j places and the phase is $e^{2\pi i\, 0.\varphi_{j+1}\varphi_{j+2}\ldots}$ — exactly the factor in the QFT product formula.
Why the other options are wrong.
(A) A phase on both branches is a global phase: it would be unobservable and is not what happens.
(C) $j\varphi$ multiplies by j instead of 2^j.
(D) $\varphi/2^j$ divides instead of multiplying by 2^j.
Remember. U^{2^j} shifts the binary point of φ by j places.

Q.29 SINGLE-CORRECT MCQ · Post-measurement state

The state $\frac{1}{\sqrt{3}}\big(\,|00\rangle + |01\rangle + |11\rangle\,\big)$ is prepared and only the *first* qubit is measured, giving 0. The probability of this outcome and the post-measurement state of the second qubit are

(A) ✓ $\frac{2}{3}$ and $|+\rangle$ **(B)** $\frac{1}{3}$ and $|0\rangle$ **(C)** $\frac{2}{3}$ and $|0\rangle$ **(D)** $\frac{1}{3}$ and $|+\rangle$

Answer: (A) **Detailed solution**

What you need to know. Group by the measured qubit, take the weight, renormalise the surviving branch.

Step-by-step solution. Group by the first qubit: $\frac{1}{\sqrt{3}}\Big(\,|0\rangle\,(\,|0\rangle + |1\rangle\,) + |1\rangle\,|1\rangle\,\Big) = \sqrt{\frac{2}{3}}\,|0\rangle\,|+\rangle + \sqrt{\frac{1}{3}}\,|1\rangle\,|1\rangle$. Probability of first qubit = 0: $\big|\sqrt{2/3}\big|^2 = \frac{2}{3}$. By the projection postulate the state collapses to the normalised $|0\rangle\,|+\rangle$, so the second qubit is $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. (Partial measurement keeps the *relative* amplitudes of the surviving branch; this is why in Shor's algorithm measuring the work register leaves the counting register in a periodic superposition.)

Why the other options are wrong.

(B) Both probability and state are wrong.

(C) $|0\rangle$: the surviving branch is $|0\rangle + |1\rangle$, not $|0\rangle$ alone.

(D) $\frac{1}{3}$ is $P(\text{first qubit} = 1)$.

Remember. Partial measurement keeps the relative amplitudes of the surviving branch.

Q.30 SINGLE-CORRECT MCQ · Unitarity of gates

Which of the following matrices can *not* be a quantum gate?

(A) ✓ $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ i & i\end{pmatrix}$ **(B)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$ **(C)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$ **(D)** $\begin{pmatrix}0 & 1\\ i & 0\end{pmatrix}$

Answer: (A) **Detailed solution**

What you need to know. Gates are unitary: orthonormal columns, $|\det| = 1$.

Step-by-step solution. Closed-system evolution is unitary: $U^\dagger U = I$, i.e. the columns are orthonormal. For $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ i & i\end{smallmatrix}\right)$ both columns equal $\frac{1}{\sqrt{2}}(1, i)^T$, so their inner product is $1 \neq 0$ (also $\det = 0$): it maps $|0\rangle$ and $|1\rangle$ to the same state and would destroy information — not unitary. Checks: H is unitary; for $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & i\\ i & 1\end{smallmatrix}\right)$ the column inner product is $\frac{1}{2}(\bar{1}\cdot i + \bar{i}\cdot 1) = \frac{1}{2}(i - i) = 0$ and each column has norm 1; $\left(\begin{smallmatrix}0 & 1\\ i & 0\end{smallmatrix}\right)$ has columns $(0, i)$ and $(1, 0)$, orthonormal. (Unitarity is also why the QFT is invertible: $F^{-1} = F^\dagger$.)

Why the other options are wrong.

(B) H is unitary (orthonormal columns).

(C) $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & i\\ i & 1\end{smallmatrix}\right)$: columns orthogonal ($\frac{1}{2}(i - i) = 0$), norms 1 — unitary.

(D) $\left(\begin{smallmatrix}0 & 1\\ i & 0\end{smallmatrix}\right)$: columns $(0, i)$, $(1, 0)$ orthonormal — unitary.

Remember. Equal columns ⇒ $\det = 0$ ⇒ not a gate.

Q.31 Single-correct MCQ · Commutation of gates

Which two-qubit gate does *not* commute with controlled-$R_2 = \mathrm{diag}(1, 1, 1, i)$?

(A) $Z \otimes Z$ **(B)** $S \otimes I$, where $S = \mathrm{diag}(1, i)$

(C) SWAP **(D)** ✓ $H \otimes I$

Answer: (D) Detailed solution

What you need to know. Diagonal matrices commute; a permutation commutes with a diagonal iff the swapped entries are equal.

Step-by-step solution. $Z \otimes Z$ and $S \otimes I$ are diagonal; diagonal matrices always commute. SWAP exchanges $|01\rangle \leftrightarrow |10\rangle$; $\mathrm{C}R_2$ has the same entry (1) on both, so $\mathrm{SWAP} \cdot \mathrm{C}R_2 \cdot \mathrm{SWAP} = \mathrm{C}R_2$: they commute (controlled-phase gates are symmetric). $H \otimes I$ mixes $|0x\rangle$ with $|1x\rangle$. For $x = 1$ the diagonal entries of $\mathrm{C}R_2$ on $|01\rangle$ and $|11\rangle$ are 1 and i — different — so the two operators do not commute. Explicitly $(H \otimes I)\,\mathrm{C}R_2\,|01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$ but $\mathrm{C}R_2(H \otimes I)\,|01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + i\,|11\rangle)$. This is why the order "H first, then the controlled rotations" in the QFT circuit matters.

Why the other options are wrong.

(A) $Z \otimes Z$ is diagonal ⇒ commutes.

(B) $S \otimes I$ is diagonal ⇒ commutes.

(C) SWAP exchanges $|01\rangle, |10\rangle$, both with entry 1 ⇒ commutes.

Remember. Order matters only for gates that mix states with different phases.

Q.32 Single-correct MCQ · State-vector memory cost

A state-vector simulator stores each amplitude as a single-precision complex number (two 4-byte floats). The memory needed for the full state of 32 qubits is

(A) 256 bytes **(B)** ✓ 32 GiB **(C)** 16 GiB **(D)** 64 GiB

Answer: (B) Detailed solution

What you need to know. Memory = $2^n \times$ bytes per amplitude; single-precision complex = 8 bytes.

Step-by-step solution. n qubits ⇒ 2^n amplitudes. Each amplitude occupies $2 \times 4 = 8$ bytes. Memory = $2^{32} \times 8 = 2^{35}$ bytes = $2^5 \times 2^{30}$ bytes = 32 GiB. Double precision (16 bytes) would need 64 GiB; 256 bytes (32×8) is the wrong idea that each qubit needs one number. Every extra qubit doubles the memory — the reason the course notebooks default to 15 qubits ($2^{15} \times 8$ B = 256 KiB) on a laptop.

Why the other options are wrong.

(A) 256 bytes counts one number per qubit, ignoring superposition.

(C) 16 GiB uses 4 bytes per amplitude (real only).

(D) 64 GiB is double precision (16 bytes).

Remember. Each extra qubit doubles the memory.

Q.33 Single-correct MCQ · Bell state in the X basis

$H \otimes H$ is applied to the Bell state $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$, followed by a measurement of both qubits. The probability of the outcome 01 is

(A) 0 **(B)** $\frac{1}{4}$ **(C)** ✓ $\frac{1}{2}$ **(D)** 1

Answer: (C) Detailed solution

What you need to know. Apply $H \otimes H$ term by term using signs $(-1)^{x \cdot y}$, then add.
Step-by-step solution. $H \otimes H\,|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ and $H \otimes H\,|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$ (sign $(-1)^{x \cdot y}$). Subtract and divide by $\sqrt{2}$:

$$\frac{1}{2\sqrt{2}}(2\,|01\rangle + 2\,|10\rangle) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$

Hence $P(01) = P(10) = \frac{1}{2}$ and $P(00) = P(11) = 0$. The perfect "equal bits" correlation in the Z basis became a perfect "opposite bits" correlation in the X basis — interference of four amplitudes, not a product-state result ($\frac{1}{4}$).
Why the other options are wrong.
(A) 0 assumes the Bell correlation (equal bits) survives in the new basis.
(B) $\frac{1}{4}$ assumes a uniform distribution (product-state thinking).
(D) 1 would need all weight on $|01\rangle$; it is shared with $|10\rangle$.
Remember. Entangled correlations change form, not strength, in another basis.

Q.34 Single-correct MCQ · Mixture vs. superposition

A source emits a qubit that is $|0\rangle$ with probability $\frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$ (a classical coin decides; nobody knows which). A Hadamard gate is applied and the qubit is measured. The probability of outcome 0 is

(A) 1 **(B)** ✓ $\frac{1}{2}$ **(C)** $\frac{1}{4}$ **(D)** 0

Answer: (B) Detailed solution

What you need to know. A mixture is averaged over cases; amplitudes of different cases never interfere.
Step-by-step solution. This is a *mixture*, not a superposition. Treat the two cases separately and average: $H\,|0\rangle = |+\rangle$ gives $P(0) = \frac{1}{2}$; $H\,|1\rangle = |-\rangle$ gives $P(0) = \frac{1}{2}$. Hence $P(0) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$. (Density matrix: $\rho = \frac{1}{2}I$, and $H\rho H^\dagger = \frac{1}{2}I$.) Contrast: the superposition $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives $H\,|+\rangle = |0\rangle$ and $P(0) = 1$. Amplitudes interfere, classical probabilities do not — QFT peaks need coherence, which is why decoherence (dephasing) destroys them.
Why the other options are wrong.
(A) 1 treats the mixture as the superposition $|+\rangle$.
(C) $\frac{1}{4}$ multiplies probabilities instead of averaging.
(D) 0 treats it as $|-\rangle$.
Remember. Interference requires coherence; decoherence kills QFT peaks.

Q.35 MATCHING LISTS · Single-qubit gate actions

Match each single-qubit gate in List-I with its action in List-II. ($R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$.)

List-I		List-II	
(P)	X	**(1)**	$\lvert 1\rangle \mapsto \frac{1}{\sqrt{2}}(\lvert 0\rangle - \lvert 1\rangle)$
(Q)	H	**(2)**	$\lvert 1\rangle \mapsto e^{i\pi/4}\lvert 1\rangle$
(R)	R_2	**(3)**	$\lvert 0\rangle \leftrightarrow \lvert 1\rangle$
(S)	R_3	**(4)**	$\lvert 1\rangle \mapsto i\lvert 1\rangle$

(A) P→3, Q→1, R→2, S→4
(B) P→1, Q→3, R→4, S→2
(C) ✓ P→3, Q→1, R→4, S→2
(D) P→4, Q→1, R→3, S→2

Answer: (C) Detailed solution

What you need to know. X flips, $H\lvert 1\rangle = \lvert -\rangle$, $R_2 = S$ ($\lvert 1\rangle \to i\lvert 1\rangle$), $R_3 = T$ ($\lvert 1\rangle \to e^{i\pi/4}\lvert 1\rangle$).
Step-by-step solution. X is the bit flip: $\lvert 0\rangle \leftrightarrow \lvert 1\rangle$ (3). $H\lvert 1\rangle = \frac{1}{\sqrt{2}}(\lvert 0\rangle - \lvert 1\rangle) = \lvert -\rangle$ (1). $R_2 = \mathrm{diag}(1, e^{2\pi i/4}) = \mathrm{diag}(1, i) = S$, so $\lvert 1\rangle \mapsto i\lvert 1\rangle$ (4). $R_3 = \mathrm{diag}(1, e^{2\pi i/8}) = T$, so $\lvert 1\rangle \mapsto e^{i\pi/4}\lvert 1\rangle$ (2). Hence P→3, Q→1, R→4, S→2. Note $R_1 = Z$, $R_2 = S$, $R_3 = T$, and $R_k^2 = R_{k-1}$.
Why the other options are wrong.
(A) R→2, S→4 swaps S and T: R_2 is the quarter turn i.
(B) P→1, Q→3 swaps X and H.
(D) P→4 says X is a phase gate.
Remember. R_k is a rotation by $360°/2^k$.

Q.36 MATCHING LISTS · Two-qubit correlations

Match each two-qubit state in List-I with its statistics when *both* qubits are measured in the computational basis (List-II).

List-I		List-II	
(P)	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle)$	**(1)**	all four outcomes, each with probability $\frac{1}{4}$
(Q)	$\frac{1}{\sqrt{2}}(\lvert 01\rangle - \lvert 10\rangle)$	**(2)**	the two bits are always equal; 00 and 11 each $\frac{1}{2}$
(R)	$\frac{1}{2}(\lvert 0\rangle + i\lvert 1\rangle)(\lvert 0\rangle - \lvert 1\rangle)$	**(3)**	first bit always 0; second bit 0 or 1 with $\frac{1}{2}$ each
(S)	$\lvert 0\rangle\lvert +\rangle$	**(4)**	the two bits always differ; 01 and 10 each $\frac{1}{2}$

(A) P→1, Q→4, R→2, S→3
(B) P→4, Q→2, R→1, S→3
(C) P→2, Q→4, R→3, S→1
(D) ✓ P→2, Q→4, R→1, S→3

Answer: (D) Detailed solution

What you need to know. Born rule on each listed state; phases do not affect computational-basis statistics.
Step-by-step solution. P (Φ^+): amplitudes only on $\lvert 00\rangle$, $\lvert 11\rangle$, each $\frac{1}{\sqrt{2}}$ ⇒ equal bits, $\frac{1}{2}$ each (2). Q (Ψ^-): only $\lvert 01\rangle$, $\lvert 10\rangle$ ⇒ bits always differ (4); the minus sign is invisible in this basis. R: expand $\frac{1}{2}(\lvert 00\rangle - \lvert 01\rangle + i\lvert 10\rangle - i\lvert 11\rangle)$; every amplitude has modulus $\frac{1}{2}$ ⇒ all four outcomes with probability $\frac{1}{4}$ (1). The phases i and -1 do not show up — just like the output of a QFT on a basis state. S: $\lvert 0\rangle\lvert +\rangle = \frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 01\rangle)$ ⇒ first bit 0, second random (3). Hence P→2, Q→4, R→1, S→3.
Why the other options are wrong.
(A) P→1 says Φ^+ gives all four outcomes; it has only 00 and 11.

(B) P→4, Q→2 swaps Φ^+ (equal bits) and Ψ^- (different bits).

(C) R→3, S→1 swaps the uniform product and $|0\rangle|+\rangle$.

Remember. Read off which basis states have non-zero amplitude.

Q.37 Assertion–Reason · Distinguishability; relative phase

Assertion (A): The states $|+\rangle$ and $|-\rangle$ cannot be distinguished by a measurement in the computational basis $\{|0\rangle, |1\rangle\}$.

Reason (R): $|+\rangle$ and $|-\rangle$ differ only by a global phase.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) ✓ A is true, but R is false

(D) A is false, but R is true

Answer: (C) Detailed solution

What you need to know. $|\pm\rangle$ differ by a relative phase; same statistics in Z, opposite in X.

Step-by-step solution. **A is true:** for both states $P(0) = P(1) = \frac{1}{2}$, because $|\pm\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$. **R is false:** $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ is not $e^{i\gamma}|+\rangle$ for any γ; the two differ by a *relative* phase -1 on $|1\rangle$. They are orthogonal ($\langle+|-\rangle = 0$) and therefore perfectly distinguishable in the X basis (apply H, then measure: $|+\rangle \to |0\rangle$, $|-\rangle \to |1\rangle$). States that differ by a global phase could never be distinguished in *any* basis. Answer: (C).

Why the other options are wrong.

(A) needs A true, R true *and* R explaining A; R is false.

(B) needs A and R both true but R not explaining A; R is false.

(D) needs A false and R true; A is true, R is false.

Remember. Indistinguishable in one basis ≠ identical.

Q.38 Statement I / II · Product-state criterion

Statement I: The Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ can be written as $|a\rangle \otimes |b\rangle$ for suitable single-qubit states $|a\rangle$, $|b\rangle$.

Statement II: A two-qubit state $c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$ is a product state if and only if $c_{00}c_{11} = c_{01}c_{10}$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

What you need to know. Rank-1 coefficient matrix ⇔ product state.

Step-by-step solution. **II is true.** ($\Rightarrow$) If the state is $(a_0|0\rangle + a_1|1\rangle)(b_0|0\rangle + b_1|1\rangle)$ then $c_{jk} = a_j b_k$ and $c_{00}c_{11} = a_0 b_0 a_1 b_1 = c_{01}c_{10}$. ($\Leftarrow$) If $c_{00}c_{11} = c_{01}c_{10}$, the matrix $C = \left(\begin{smallmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{smallmatrix}\right)$ has $\det C = 0$, so it has rank 1 and $C = \mathbf{a}\mathbf{b}^T$, i.e. $c_{jk} = a_j b_k$. **I is false:** for Φ^+, $c_{00}c_{11} = \frac{1}{2} \neq 0 = c_{01}c_{10}$. Answer: (D).

Why the other options are wrong.

(A) needs both statements true; I is false.

(B) needs both statements false; II is true.

(C) needs I true and II false; the truth values are the other way round.

Remember. $\det C = 0$ is the two-qubit entanglement test.

Paragraph for Questions 39 and 40

Hadamard test. A control qubit starts in $|0\rangle$ and a target register starts in an eigenstate $|u\rangle$ of a unitary U with $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $0 \le \varphi < 1$. The circuit applies, in order: (i) H on the control; (ii) controlled-U (control qubit → target register); (iii) H on the control; (iv) a measurement of the control qubit in the computational basis. This is the one-counting-qubit version of quantum phase estimation.

Q.39 PARAGRAPH BASED · Hadamard test

The probability that the control qubit is found in state $|1\rangle$ is

(A) $\cos^2(\pi\varphi)$ **(B)** $\sin^2(\pi\varphi/2)$ **(C)** $\sin^2(2\pi\varphi)$ **(D)** ✓ $\sin^2(\pi\varphi)$

Answer: (D) Detailed solution

What you need to know. Hadamard test = H, kickback, H: $P(1) = \sin^2(\pi\varphi)$.
Step-by-step solution. After (i): $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)|u\rangle$. After (ii), by phase kickback (Q.28 with $j=0$): $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\varphi}|1\rangle)|u\rangle$. After (iii) (Q.23 with $\phi = 2\pi\varphi$):

$$\Big(\frac{1+e^{2\pi i\varphi}}{2}|0\rangle + \frac{1-e^{2\pi i\varphi}}{2}|1\rangle\Big)|u\rangle\,.$$

$|1-e^{2\pi i\varphi}|^2 = 2-2\cos(2\pi\varphi) = 4\sin^2(\pi\varphi)$, so $P(1) = \sin^2(\pi\varphi)$ and $P(0) = \cos^2(\pi\varphi)$. $\varphi = 0 \Rightarrow$ always 0; $\varphi = \frac{1}{2} \Rightarrow$ always 1: one qubit resolves exactly one binary digit $\varphi = 0.\varphi_1$. More digits need more counting qubits and an inverse QFT.
Why the other options are wrong.
(A) $\cos^2(\pi\varphi)$ is $P(0)$.
(B) $\sin^2(\pi\varphi/2)$ halves the angle twice.
(C) $\sin^2(2\pi\varphi)$ uses the full angle instead of the half angle.
Remember. One counting qubit resolves one binary digit of φ.

Q.40 PARAGRAPH BASED · Repetition and confidence

Take $\varphi = \frac{1}{3}$. The circuit is run repeatedly (fresh $|0\rangle$ control each time). The minimum number of runs needed so that the probability of observing outcome 1 *at least once* exceeds 0.99 is

(A) 2 **(B)** 3 **(C)** ✓ 4 **(D)** 7

Answer: (C) Detailed solution

What you need to know. Independent repetitions: $P(\text{at least one success}) = 1-(1-p)^k$.
Step-by-step solution. From Q.39, $P(1) = \sin^2(\pi/3) = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$, so a single run gives 0 with probability $\frac{1}{4}$. Runs are independent: $P(\text{no 1 in } k \text{ runs}) = \left(\frac{1}{4}\right)^k$. We need $1-4^{-k} > 0.99$, i.e. $4^k > 100$. $4^3 = 64 < 100$ (failure probability 0.0156), $4^4 = 256 > 100$ (failure probability 0.0039). Hence $k = 4$. (Repetition amplifies confidence; it does not make the single-shot estimate of $\varphi = \frac{1}{3} = 0.0101\ldots_2$ exact.)
Why the other options are wrong.
(A) 2 runs give failure 0.0625.
(B) 3 runs give failure $4^{-3} = 0.0156 > 0.01$.

(D) 7 is the answer for $p = \frac{1}{2}$, not $p = \frac{3}{4}$.
Remember. Find p first, then the smallest k with $(1-p)^k < \delta$.

SILICO FELLER

SECTION C · QFT Basics Q.41–60

Q.41 Single-correct MCQ · QFT matrix element ($n = 2$)

For $n = 2$, the amplitude of $|10\rangle$ in $\mathrm{QFT}\,|11\rangle$ is

(A) ✓ $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** $-\frac{i}{2}$ **(D)** $\frac{i}{2}$

Answer: (A) **Detailed solution**

What you need to know. Amplitude $\langle y|\mathrm{QFT}|x\rangle = \omega^{xy}/\sqrt{N}$; convert bit strings to integers first.
Step-by-step solution. $N = 4$, $\omega = e^{2\pi i/4} = i$, $x = |11\rangle = 3$, $y = |10\rangle = 2$. From the definition

$$\langle y|\mathrm{QFT}|x\rangle = \frac{1}{\sqrt{N}}\,\omega^{xy} = \frac{1}{2}\,i^6 = \frac{1}{2}\,(i^4)(i^2) = -\frac{1}{2}.$$

Full output: $\mathrm{QFT}\,|3\rangle = \frac{1}{2}(\,|0\rangle - i\,|1\rangle - |2\rangle + i\,|3\rangle\,)$, i.e. $\frac{1}{2}(|00\rangle - i\,|01\rangle - |10\rangle + i\,|11\rangle)$.
Why the other options are wrong.
(B) $\frac{1}{2}$ drops the sign of $i^6 = -1$.
(C) $-\frac{i}{2}$ uses $xy = 3$ (wrong integers).
(D) $\frac{i}{2}$ uses i^{xy} with $xy = 1$ or 5 (wrong integers).
Remember. $|10\rangle = 2$, $|11\rangle = 3$; reduce xy mod N.

Q.42 Single-correct MCQ · Output distribution of QFT $|x\rangle$

A 5-qubit QFT is applied to the basis state $|x\rangle$ with $x = 19$ and the output is measured. The probability of obtaining $y = 7$ is

(A) $\frac{7}{32}$ **(B)** $\frac{1}{\sqrt{32}}$ **(C)** 0 **(D)** ✓ $\frac{1}{32}$

Answer: (D) **Detailed solution**

What you need to know. $|\omega^{xy}| = 1$, so every outcome has probability $1/N$ for a basis input.
Step-by-step solution. $\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{32}}\sum_y \omega^{xy}\,|y\rangle$ and $|\omega^{xy}| = 1$, so every outcome has probability $\left|\frac{1}{\sqrt{32}}\right|^2 = \frac{1}{32}$, whatever x and y are. ($\frac{1}{\sqrt{32}}$ is the modulus of the amplitude, not a probability.) All the information about x sits in the phases ω^{19y}.
Why the other options are wrong.
(A) $\frac{7}{32}$ invents a dependence on y.
(B) $\frac{1}{\sqrt{32}}$ is the modulus of the amplitude, not the probability.
(C) 0: no outcome is forbidden for a basis input.
Remember. QFT of a basis state = flat distribution, information in phases.

Q.43 Single-correct MCQ · Product formula ($n = 3$)

For $n = 3$, QFT $|110\rangle$ equals (qubit 1 written first)

(A) $\frac{1}{\sqrt{8}}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$ **(B)** $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle + i|1\rangle)$

(C) ✓ $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$ **(D)** $\frac{1}{\sqrt{8}}(|0\rangle - i|1\rangle)(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$

Answer: (C) Detailed solution

What you need to know. Product formula: output qubit l carries $0.x_{n-l+1}\dots x_n$.

Step-by-step solution. *Product formula.* Write $y = \sum_{l=1}^{n} y_l 2^{n-l}$; then $\omega^{xy} = \prod_l e^{2\pi i x y_l/2^l}$ and $e^{2\pi i\, x/2^l} = e^{2\pi i\, 0.x_{n-l+1}\dots x_n}$ (only the last l bits of x survive), so the sum over y factorises into one factor per output qubit. For $n = 3$: QFT $|x_1x_2x_3\rangle = \frac{1}{\sqrt{8}}(|0\rangle + e^{2\pi i\, 0.x_3}|1\rangle)(|0\rangle + e^{2\pi i\, 0.x_2x_3}|1\rangle)(|0\rangle + e^{2\pi i\, 0.x_1x_2x_3}|1\rangle)$. With $x_1x_2x_3 = 110$: $0.x_3 = 0.0 = 0 \Rightarrow e^0 = 1$; $0.x_2x_3 = 0.10_2 = \frac{1}{2} \Rightarrow e^{i\pi} = -1$; $0.x_1x_2x_3 = 0.110_2 = \frac{3}{4} \Rightarrow e^{3\pi i/2} = -i$. Hence $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$. *Check* with the definition: amplitude of y is $\omega^{6y}/\sqrt{8} = (-i)^y/\sqrt{8}$; for $y = 4y_1 + 2y_2 + y_3$, $(-i)^{4y_1}(-i)^{2y_2}(-i)^{y_3} = 1 \cdot (-1)^{y_2}(-i)^{y_3}$ — the same three factors. The option $\frac{1}{\sqrt{8}}(|0\rangle - i|1\rangle)(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$ is the bit-reversed order (what you get without the final SWAPs).

Why the other options are wrong.

(A) $(|0\rangle - |1\rangle)$ in the first factor uses $0.x_3 = \frac{1}{2}$, but $x_3 = 0$.

(B) $(|0\rangle + i|1\rangle)$ in the last factor uses the wrong sign ($0.110 = \frac{3}{4}$ gives $-i$).

(D) Reversed factor order: the state before the final SWAPs.

Remember. Check one amplitude with the definition after using the product formula.

Q.44 Single-correct MCQ · Controlled-rotation count

The textbook QFT circuit on $n = 12$ qubits contains how many controlled-R_k gates?

(A) 144 **(B)** 72 **(C)** ✓ 66 **(D)** 78

Answer: (C) Detailed solution

What you need to know. Qubit j receives $n - j$ controlled rotations; total $\binom{n}{2}$.

Step-by-step solution. Qubit j receives one controlled rotation from each qubit below it, i.e. $n-j$ of them. Total

$$\sum_{j=1}^{n}(n-j) = (n-1) + (n-2) + \cdots + 0 = \frac{n(n-1)}{2} = \frac{12 \cdot 11}{2} = 66.$$

($78 = \frac{n(n+1)}{2}$ also counts the 12 Hadamards; $144 = n^2$.)

Why the other options are wrong.

(A) $144 = n^2$ overcounts.

(B) 72 is not a formula of the circuit.

(D) $78 = \frac{n(n+1)}{2}$ also counts the n Hadamards.

Remember. $\frac{n(n-1)}{2}$ controlled rotations.

Q.45 SINGLE-CORRECT MCQ · Two-qubit gates incl. SWAPs

On hardware a SWAP is implemented with 3 CNOTs. Counting every controlled-R_k as one two-qubit gate, the total number of two-qubit gates in the 12-qubit textbook QFT (including the final SWAPs) is

(A) 72 **(B)** ✓ 84 **(C)** 102 **(D)** 66

Answer: (B) Detailed solution

What you need to know. Two-qubit gates = controlled rotations + 3 per SWAP; $\lfloor n/2 \rfloor$ SWAPs.
Step-by-step solution. Controlled rotations: $\frac{12\cdot 11}{2} = 66$ (Q.44). SWAPs: $\lfloor 12/2 \rfloor = 6$, each costing 3 CNOTs = 18. Total $66+18 = 84$. ($72 = 66+6$ counts each SWAP as one gate; $102 = 66+36$ uses 6 CNOTs per SWAP.) In practice the SWAPs are often removed by simply relabelling the output qubits, leaving 66.
Why the other options are wrong.
(A) 72 counts each SWAP as one gate.
(C) 102 uses 6 CNOTs per SWAP or 12 SWAPs.
(D) 66 forgets the SWAPs.
Remember. $\binom{n}{2} + 3\lfloor n/2 \rfloor$.

Q.46 SINGLE-CORRECT MCQ · Smallest rotation angle

The smallest rotation angle appearing in the textbook QFT circuit on 6 qubits is

(A) 11.25° **(B)** 6° **(C)** 2.8125° **(D)** ✓ 5.625°

Answer: (D) Detailed solution

What you need to know. The smallest rotation is R_n with angle $360°/2^n$.
Step-by-step solution. The rotations used are $R_2, \ldots, R_n$ with angle $\frac{2\pi}{2^k}$. The smallest is $R_n = R_6$: $\frac{360°}{2^6} = \frac{360°}{64} = 5.625°$. (11.25° is R_5; 2.8125° would be R_7, which needs 7 qubits.) These tiny angles are why the approximate QFT drops the smallest rotations (Section D).
Why the other options are wrong.
(A) 11.25° is R_5.
(B) 6° divides 360° by $n \cdot 10$ — not a QFT angle.
(C) 2.8125° is R_7, which needs 7 qubits.
Remember. Angles halve with each extra qubit.

Q.47 SINGLE-CORRECT MCQ · Phase on qubit 1; role of SWAPs

In the textbook QFT circuit acting on $|x_1 x_2 \ldots x_n\rangle$, *before* the final SWAPs, qubit 1 is in the state $\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\,\phi}|1\rangle\right)$ with ϕ equal to

(A) ✓ $0.x_1x_2\ldots x_n$ **(B)** $0.x_nx_{n-1}\ldots x_1$ **(C)** $0.x_n$ **(D)** $0.x_1$

Answer: (A) Detailed solution

What you need to know. H gives $0.x_1$; each controlled R_k appends bit x_k.
Step-by-step solution. H on qubit 1 gives $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\,0.x_1}|1\rangle)$ because $(-1)^{x_1} = e^{2\pi i\,x_1/2}$. Each controlled-$R_k$ from qubit k (for $k = 2, \ldots, n$) multiplies the $|1\rangle$ part by $e^{2\pi i\,x_k/2^k}$, appending the binary digit x_k at position k: $0.x_1 \to 0.x_1x_2 \to \cdots \to 0.x_1x_2\ldots x_n$. This factor belongs to the *last* output qubit in the product formula, so the SWAPs move it from qubit 1 to qubit n.
Why the other options are wrong.

(B) $0.x_n \dots x_1$ reads the bits in reverse.

(C) $0.x_n$ is the phase of qubit n before the SWAPs.

(D) $0.x_1$ is the phase right after H, before the rotations.

Remember. Before the SWAPs, qubit j holds $0.x_j \dots x_n$.

Q.48 SINGLE-CORRECT MCQ · QFT^2 = negation

For any n-qubit basis state, $\mathrm{QFT}^2 |x\rangle$ equals ($N = 2^n$)

(A) $|x\rangle$ **(B)** $|2x \bmod N\rangle$ **(C)** ✓ $|(-x) \bmod N\rangle$ **(D)** $|N-1-x\rangle$

Answer: (C) Detailed solution

What you need to know. $\mathrm{QFT}^2 |x\rangle = \frac{1}{N} \sum_z \sum_y \omega^{y(x+z)} |z\rangle$; use orthogonality.

Step-by-step solution.

$$\mathrm{QFT}^2 |x\rangle = \frac{1}{N} \sum_z \Big(\sum_y \omega^{y(x+z)} \Big) |z\rangle .$$

The inner geometric sum equals N if $x + z \equiv 0 \pmod N$ and 0 otherwise. Hence $\mathrm{QFT}^2 |x\rangle = |-x \bmod N\rangle$. Examples ($n = 2$): $|0\rangle \to |0\rangle$, $|1\rangle \to |3\rangle$, $|2\rangle \to |2\rangle$. Consequently $\mathrm{QFT}^4 = I$. ($|N-1-x\rangle$ is the bitwise complement, a different map: it sends $|0\rangle \to |N-1\rangle$.)

Why the other options are wrong.

(A) $|x\rangle$ would mean $\mathrm{QFT}^2 = I$; that is only true for QFT^4.

(B) $|2x\rangle$ adds the exponents wrongly.

(D) $|N-1-x\rangle$ is the bit complement, a different map ($0 \to N-1$).

Remember. QFT^2 is the reflection $x \mapsto -x$; $\mathrm{QFT}^4 = I$.

Q.49 SINGLE-CORRECT MCQ · QFT of the uniform state

The QFT is applied to the uniform superposition $\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$. The output is

(A) $|N-1\rangle$ **(B)** ✓ $|0\rangle$

(C) the uniform superposition itself **(D)** $|N/2\rangle$

Answer: (B) Detailed solution

What you need to know. $\sum_x \omega^{xy} = N\delta_{y0}$: a constant signal has only zero frequency.

Step-by-step solution.

$$\frac{1}{\sqrt{N}} \sum_x \frac{1}{\sqrt{N}} \sum_y \omega^{xy} |y\rangle = \frac{1}{N} \sum_y \Big(\sum_x \omega^{xy} \Big) |y\rangle = |0\rangle ,$$

since $\sum_x \omega^{xy} = N\,\delta_{y,0}$. A constant ("DC") signal has only the zero-frequency component. Alternatively: the uniform superposition is $\mathrm{QFT}\,|0\rangle$ (product formula with $x = 0$), so applying the QFT once more gives $\mathrm{QFT}^2 |0\rangle = |-0 \bmod N\rangle = |0\rangle$ by Q.48.

Why the other options are wrong.

(A) $|N-1\rangle$ confuses the output with the bit complement.

(C) The uniform state is the QFT of $|0\rangle$, not its own QFT.

(D) $|N/2\rangle$ is the highest frequency, produced by an alternating input.

Remember. Uniform $\leftrightarrow |0\rangle$ under the QFT.

Q.50 Single-correct MCQ · QFT of $|10\ldots0\rangle$

For n qubits, $\mathrm{QFT}\,|10\ldots0\rangle$ (i.e. $x = 2^{n-1}$) equals

(A) ✓ $|+\rangle^{\otimes(n-1)} \otimes |-\rangle$ **(B)** $|-\rangle^{\otimes n}$ **(C)** $|+\rangle^{\otimes n}$ **(D)** $|-\rangle \otimes |+\rangle^{\otimes(n-1)}$

Answer: (A) Detailed solution

What you need to know. $\omega^{2^{n-1}y} = (-1)^y$ depends only on the last bit of y.
Step-by-step solution. Amplitude of y: $\frac{1}{\sqrt{N}}\omega^{2^{n-1}y} = \frac{1}{\sqrt{N}}e^{2\pi i\, y/2} = \frac{1}{\sqrt{N}}(-1)^y$. The sign depends only on the parity of y, i.e. on its last bit y_n (qubit n). So the state factorises as $|+\rangle\otimes\cdots\otimes|+\rangle\otimes|-\rangle$. Product-formula check: $x_1 = 1$, others 0, so every factor has phase $e^{2\pi i\, 0.0\ldots} = 1$ except the last, $e^{2\pi i\, 0.x_1\ldots x_n} = e^{2\pi i\, 0.1} = -1$. (This is the "highest frequency" alternating signal $+,-,+,-,\ldots$.)
Why the other options are wrong.
(B) $|-\rangle^{\otimes n}$ would need signs depending on every bit.
(C) $|+\rangle^{\otimes n}$ is $\mathrm{QFT}\,|0\rangle$.
(D) $|-\rangle$ on qubit 1 puts the sign on the most significant bit.
Remember. The highest bit of x only affects the last output qubit.

Q.51 Single-correct MCQ · Constructing the inverse QFT

The inverse QFT on 3 qubits is built by reversing the textbook QFT circuit and conjugating every gate. Ignoring the SWAP, the *first* gate of the inverse circuit is

(A) ✓ H on qubit 3
(B) controlled-$R_3^\dagger$ between qubits 1 and 3
(C) controlled-$R_2^\dagger$ between qubits 1 and 2
(D) H on qubit 1

Answer: (A) Detailed solution

What you need to know. Inverse circuit: reverse the order, invert each gate.
Step-by-step solution. $\mathrm{QFT} = \mathrm{SWAP}_{13}\cdot H_3 \cdot \mathrm{CR}_2^{(3\to2)} \cdot H_2 \cdot \mathrm{CR}_3^{(3\to1)} \cdot \mathrm{CR}_2^{(2\to1)} \cdot H_1$ (rightmost acts first). Since $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$:

$$\mathrm{QFT}^{-1} = H_1 \cdot \mathrm{CR}_2^{\dagger\,(2\to1)} \cdot \mathrm{CR}_3^{\dagger\,(3\to1)} \cdot H_2 \cdot \mathrm{CR}_2^{\dagger\,(3\to2)} \cdot H_3 \cdot \mathrm{SWAP}_{13},$$

so the order of application is SWAP, then H on qubit 3, then $\mathrm{CR}_2^\dagger$ (3→2), H_2, …, finally H_1. $H^\dagger = H$ and $R_k^\dagger = \mathrm{diag}(1, e^{-2\pi i/2^k})$.
Why the other options are wrong.
(B) $CR_3^\dagger$ (1,3) comes later, after H_2.
(C) $CR_2^\dagger$ (1,2) is the second-to-last gate.
(D) H on qubit 1 is the *last* gate of the inverse, not the first: the inverse starts where the QFT ended (qubit 3).
Remember. $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Q.52 SINGLE-CORRECT MCQ · Phase of a QFT amplitude

For $n = 4$, the phase (in degrees, in $[0°, 360°)$) of the amplitude of $|y\rangle$, $y = 3$, in QFT $|x\rangle$, $x = 5$, is

(A) 22.5° **(B)** ✓ 337.5° **(C)** 168.75° **(D)** 15°

Answer: (B) Detailed solution

What you need to know. Phase $= 360° \cdot \frac{xy \bmod N}{N}$.
Step-by-step solution. The amplitude is $\frac{1}{4}\omega^{xy}$ with $\omega = e^{2\pi i/16}$, so the phase is $\frac{2\pi\, xy}{16} = 2\pi \cdot \frac{15}{16}$, i.e. $\frac{15}{16} \times 360° = 337.5°$ (equivalently $-22.5°$; the option 22.5° has the sign convention reversed).
Why the other options are wrong.
(A) 22.5° uses the opposite sign ($e^{-2\pi i}$).
(C) 168.75° divides by 32 instead of 16.
(D) 15° is xy read as degrees.
Remember. Compute xy mod N, then scale to 360°.

Q.53 SINGLE-CORRECT MCQ · QFT vs. NumPy FFT conventions

NumPy's two transforms of a length-N vector a are

$$\texttt{fft}(a)_y = \sum_x a_x\, e^{-2\pi i\, xy/N}, \qquad \texttt{ifft}(a)_y = \frac{1}{N}\sum_x a_x\, e^{+2\pi i\, xy/N}.$$

If QFT $\sum_x \alpha_x |x\rangle = \sum_y \beta_y |y\rangle$ (convention box), then the vector β equals

(A) $\texttt{fft}(\alpha)$ **(B)** ✓ $\sqrt{N} \cdot \texttt{ifft}(\alpha)$ **(C)** $\texttt{ifft}(\alpha)/\sqrt{N}$ **(D)** $\texttt{fft}(\alpha)/\sqrt{N}$

Answer: (B) Detailed solution

What you need to know. Match the exponent sign and normalisation: QFT uses $+$ and $\frac{1}{\sqrt{N}}$.
Step-by-step solution. $\beta_y = \frac{1}{\sqrt{N}}\sum_x \alpha_x e^{+2\pi i\, xy/N}$. The sign of the exponent is $+$, which matches `ifft`, and $\frac{1}{\sqrt{N}} = \sqrt{N} \cdot \frac{1}{N}$, so $\beta = \sqrt{N}\,\texttt{ifft}(\alpha)$ (`numpy.fft.ifft(alpha, norm="ortho")`). $\texttt{fft}(\alpha)/\sqrt{N}$ is the *inverse* QFT. This is how the course notebooks verify CUDA-Q's state vectors.
Why the other options are wrong.
(A) `fft` has the wrong sign and no normalisation.
(C) $\texttt{ifft}/\sqrt{N}$ has normalisation $N^{-3/2}$.
(D) $\texttt{fft}/\sqrt{N}$ has the minus sign: it is the inverse QFT.
Remember. $\beta = \sqrt{N}\,\texttt{ifft}(\alpha) =$ `ifft(alpha, norm="ortho")`.

Q.54 SINGLE-CORRECT MCQ · QFT $|x\rangle$ is a product state

For an n-qubit computational basis state $|x\rangle$, the state QFT $|x\rangle$ is

(A) a product state only when x is even **(B)** always maximally entangled
(C) entangled unless $x = 0$ **(D)** ✓ always a product (unentangled) state

Answer: (D) Detailed solution

What you need to know. The product formula writes QFT $|x\rangle$ as a tensor product.
Step-by-step solution. The product formula writes QFT $|x\rangle$ as a tensor product of n single-qubit states $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x_{n-l+1}\ldots x_n}|1\rangle)$, $l = 1, \ldots, n$. Hence it is never entangled, for *every* x. (The QFT *can* entangle superposition inputs, because the controlled rotations are entangling gates — compare Q.25.) This product structure is also why the textbook circuit needs only $O(n^2)$ gates.

Why the other options are wrong.

(A) Product for every x, not only even x.

(B) Never entangled for basis inputs, so "maximally entangled" is false.

(C) Product even for $x \neq 0$.

Remember. Basis input ⇒ product output; superposition input can entangle.

Q.55 MATCHING LISTS · Gate count $n(n-1)/2$

Match the number of qubits n (List-I) with the number of controlled-R_k gates in the textbook QFT (List-II).

List-I		List-II	
(P)	$n = 4$	**(1)**	15
(Q)	$n = 6$	**(2)**	28
(R)	$n = 8$	**(3)**	45
(S)	$n = 10$	**(4)**	6

(A) P→1, Q→4, R→2, S→3

(B) P→4, Q→2, R→1, S→3

(C) ✓ P→4, Q→1, R→2, S→3

(D) P→4, Q→1, R→3, S→2

Answer: (C) Detailed solution

What you need to know. $\binom{n}{2}$ for each n.

Step-by-step solution. Count $= \binom{n}{2} = \frac{n(n-1)}{2}$: $n = 4 \to 6$ (4), $n = 6 \to 15$ (1), $n = 8 \to 28$ (2), $n = 10 \to 45$ (3). Hence P→4, Q→1, R→2, S→3. Growth is quadratic, compared with the exponential 2^n amplitudes transformed.

Why the other options are wrong.

(A) P→1 swaps $n = 4$ and $n = 6$.

(B) Q→2 assigns 28 to $n = 6$.

(D) R→3 assigns 45 to $n = 8$.

Remember. $6, 15, 28, 45$ for $n = 4, 6, 8, 10$.

Q.56 MATCHING LISTS · Two-qubit QFT outputs

For $n = 2$, match each input (List-I) with QFT of it (List-II); qubit 1 is written first.

List-I		List-II	
(P)	$\lvert 00\rangle$	**(1)**	$\frac{1}{2}(\lvert 0\rangle - \lvert 1\rangle)(\lvert 0\rangle + i\lvert 1\rangle)$
(Q)	$\lvert 01\rangle$	**(2)**	$\frac{1}{2}(\lvert 0\rangle - \lvert 1\rangle)(\lvert 0\rangle - i\lvert 1\rangle)$
(R)	$\lvert 10\rangle$	**(3)**	$\frac{1}{2}(\lvert 0\rangle + \lvert 1\rangle)(\lvert 0\rangle + \lvert 1\rangle)$
(S)	$\lvert 11\rangle$	**(4)**	$\frac{1}{2}(\lvert 0\rangle + \lvert 1\rangle)(\lvert 0\rangle - \lvert 1\rangle)$

(A) P→3, Q→4, R→1, S→2

(B) P→4, Q→1, R→3, S→2

(C) P→3, Q→2, R→4, S→1

(D) ✓ P→3, Q→1, R→4, S→2

Answer: (D) Detailed solution

What you need to know. Product formula for $n = 2$: $(|0\rangle + e^{2\pi i 0.x_2}|1\rangle)(|0\rangle + e^{2\pi i 0.x_1x_2}|1\rangle)/2$.
Step-by-step solution. Product formula for $n = 2$: $\frac{1}{2}(|0\rangle + e^{2\pi i\, 0.x_2}|1\rangle)(|0\rangle + e^{2\pi i\, 0.x_1x_2}|1\rangle)$.

- 00: phases $e^0, e^0 \Rightarrow (|0\rangle + |1\rangle)(|0\rangle + |1\rangle)/2$ (3).
- 01: $0.x_2 = \frac{1}{2} \to -1$; $0.x_1x_2 = 0.01_2 = \frac{1}{4} \to i \Rightarrow$ (1).
- 10: $0.x_2 = 0 \to 1$; $0.10_2 = \frac{1}{2} \to -1 \Rightarrow$ (4).
- 11: $0.x_2 = \frac{1}{2} \to -1$; $0.11_2 = \frac{3}{4} \to -i \Rightarrow$ (2).

Hence P→3, Q→1, R→4, S→2. Check $|11\rangle$ against Q.41: expanding (2) gives $\frac{1}{2}(|00\rangle - i|01\rangle - |10\rangle + i|11\rangle)$, whose $|10\rangle$ amplitude is $-\frac{1}{2}$.
Why the other options are wrong.
(A) Q→4, R→1 swaps $|01\rangle$ and $|10\rangle$ (bit order).
(B) P→4 says QFT $|00\rangle$ has a minus sign; it is uniform.
(C) Q→2, S→1 swaps $|01\rangle$ and $|11\rangle$ (sign of i).
Remember. Do the four rows independently with the product formula.

Q.57 Assertion–Reason · Why measuring QFT $|x\rangle$ reveals nothing

Assertion (A): Measuring QFT $|x\rangle$ in the computational basis reveals no information about the basis state x.

Reason (R): All N amplitudes of QFT $|x\rangle$ have the same magnitude $1/\sqrt{N}$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

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Answer: (A) Detailed solution

What you need to know. Equal magnitudes ⇒ uniform outcome distribution independent of x.
Step-by-step solution. **R is true:** $|\omega^{xy}/\sqrt{N}| = 1/\sqrt{N}$ for every y. **A is true** *because* of R: the outcome distribution $P(y) = 1/N$ is the same uniform distribution for every x, so the result is statistically independent of x. The information is entirely in the relative phases, which a computational-basis measurement discards; one must first undo the QFT (or interfere the phases in some other way) to read it. R is the correct explanation. Answer: (A).
Why the other options are wrong.
(B) needs A and R both true but R not explaining A; here R is exactly the reason.
(C) needs A true and R false; R is true.
(D) needs A false and R true; A is true.
Remember. Information hidden in phases needs interference to be read.

Q.58 Statement I / II · What the QFT does *not* speed up

Statement I: The QFT on n qubits can be implemented with $O(n^2)$ one- and two-qubit gates.

Statement II: Consequently, for an arbitrary classical input vector of length 2^n, a quantum computer can output all 2^n of its Fourier coefficients using only $O(n^2)$ operations in total.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

What you need to know. Gate count vs input loading and output readout.

Step-by-step solution. **I is true:** n Hadamards, $\frac{n(n-1)}{2}$ controlled rotations and $\lfloor n/2 \rfloor$ SWAPs. **II is false**, for two reasons: (1) *input* — loading an arbitrary 2^n-entry vector into amplitudes generally needs $\Omega(2^n)$ gates; (2) *output* — the coefficients are amplitudes; a measurement returns just one index y with probability $|\beta_y|^2$, and estimating all 2^n values needs exponentially many repetitions. The QFT is useful as a *subroutine* (phase estimation, Shor) where only a few peaks need to be read. Answer: (C).

Why the other options are wrong.

(A) needs both statements true; II is false.

(B) needs both statements false; I is true.

(D) needs I false and II true; I is true.

Remember. The QFT is a subroutine, not a fast classical FFT.

Paragraph for Questions 59 and 60

The textbook 3-qubit QFT circuit is applied to the input $|x_1x_2x_3\rangle = |011\rangle$ ($x = 3$). The gates, in order, are: H on qubit 1; controlled-R_2 (control qubit 2, target qubit 1); controlled-R_3 (control qubit 3, target qubit 1); H on qubit 2; controlled-R_2 (control qubit 3, target qubit 2); H on qubit 3; SWAP of qubits 1 and 3. Here $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, and "the phase of qubit j" means the angle θ in its state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$, taken in $[0°, 360°)$.

Q.59 Paragraph based · Circuit trace: phase build-up

Immediately after the controlled-R_3 gate (the third gate), the phase of qubit 1 is

(A) 270° **(B)** ✓ 135° **(C)** 90° **(D)** 45°

Answer: (B) Detailed solution

What you need to know. Gate-by-gate phase build-up on qubit 1: $180°x_1 + 90°x_2 + 45°x_3$.

Step-by-step solution. Input $x_1 = 0$, $x_2 = 1$, $x_3 = 1$. H on qubit 1: phase $2\pi\, 0.x_1 = 0$, state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. C-$R_2$ (control $x_2 = 1$): adds $\frac{2\pi}{4} = 90°$. C-R_3 (control $x_3 = 1$): adds $\frac{2\pi}{8} = 45°$. Total $0° + 90° + 45° = 135°$, i.e. $2\pi \cdot 0.011_2 = 2\pi \cdot \frac{3}{8}$. (Since qubits 2 and 3 are basis states, the controls are classical here and nothing becomes entangled.)

Why the other options are wrong.

(A) 270° treats $x_1 = 1$.

(C) 90° stops after CR_2.

(D) 45° counts only CR_3.

Remember. Each controlled rotation adds its angle only if its control bit is 1.

Q.60 PARAGRAPH BASED · Circuit trace: final amplitude, SWAP

After the complete circuit (including the SWAP), the amplitude of the basis state $|110\rangle$ is

(A) $\frac{i}{8}$ **(B)** $\frac{e^{i\pi/4}}{\sqrt{8}}$ **(C)** $\frac{-i}{\sqrt{8}}$ **(D)✓** $\frac{i}{\sqrt{8}}$

Answer: (D) Detailed solution

What you need to know. With the SWAP the circuit equals the QFT; amplitude $= \omega^{xy}/\sqrt{N}$.
Step-by-step solution. With the SWAP, the circuit implements exactly QFT $|3\rangle = \frac{1}{\sqrt{8}}\sum_y \omega^{3y}|y\rangle$, $\omega = e^{i\pi/4}$. For $|110\rangle$, $y = 6$: $\omega^{18} = \omega^{16}\omega^2 = e^{i\pi/2} = i$, so the amplitude is $\frac{i}{\sqrt{8}}$. Gate-by-gate check: before the SWAP, qubit 1 has phase $\frac{3}{8}$ turn (Q.59), qubit 2 has $2\pi\, 0.x_2x_3 = 2\pi\, 0.11_2 = 270°$, qubit 3 has $2\pi\, 0.x_3 = 180°$. After swapping qubits 1 and 3, $|110\rangle$ collects phases $180° + 270° = 450° \equiv 90°$ $\Rightarrow i$. Forgetting the SWAP gives $\frac{3}{8} + \frac{3}{4} = \frac{9}{8} \equiv \frac{1}{8}$ turn, i.e. $e^{i\pi/4}/\sqrt{8}$ — a wrong option.
Why the other options are wrong.
(A) $i/8$ uses the wrong normalisation ($1/8$ instead of $1/\sqrt{8}$).
(B) $e^{i\pi/4}/\sqrt{8}$ is what you get without the SWAP.
(C) $-i/\sqrt{8}$ uses $e^{-2\pi i}$.
Remember. Always include the final SWAPs (or relabel the output).

SILICO FELLER

SECTION D · Advanced QFT Q.61–80

Q.61 Single-correct MCQ · QFT of a periodic state $(r \mid N)$

On $n = 6$ qubits ($N = 64$) the state $\frac{1}{\sqrt{8}}\sum_{j=0}^{7}|3+8j\rangle$ is prepared, the QFT is applied and all qubits are measured. The possible outcomes y and their probabilities are

(A) only $y = 8$, with probability 1

(B) all 64 values of y, each with probability $\frac{1}{64}$

(C) $y \in \{3, 11, 19, \ldots, 59\}$, each with probability $\frac{1}{8}$

(D) ✓ $y \in \{0, 8, 16, \ldots, 56\}$, each with probability $\frac{1}{8}$

Answer: (D) Detailed solution

What you need to know. Period $r \mid N \Rightarrow r$ peaks at multiples of N/r, probability $\frac{1}{r}$ each.

Step-by-step solution. With $\omega = e^{2\pi i/64}$,

$$\beta_y = \frac{1}{\sqrt{8}} \cdot \frac{1}{\sqrt{64}} \sum_{j=0}^{7} \omega^{(3+8j)y} = \frac{\omega^{3y}}{\sqrt{512}} \sum_{j=0}^{7} e^{2\pi i\, jy/8}.$$

The inner sum is 8 if $8 \mid y$ and 0 otherwise. Hence $\beta_y = \frac{8\omega^{3y}}{\sqrt{512}} = \frac{\omega^{3y}}{\sqrt{8}}$ for $y = 0, 8, \ldots, 56$ (8 values, probability $\frac{1}{8}$ each) and $\beta_y = 0$ elsewhere. A state with period r (here 8) dividing N produces r peaks at the multiples of N/r; the offset 3 only changes phases (see Q.63).

Why the other options are wrong.

(A) A single peak would require a constant (period-1) input.

(B) A flat distribution is the QFT of a single basis state.

(C) Peaks at $3, 11, \ldots$ copy the input positions; the offset moves into phases, not positions.

Remember. Period in $x \leftrightarrow$ spacing N/r in y.

Q.62 Single-correct MCQ · Phases of QFT peaks

For the state $\frac{1}{\sqrt{8}}\sum_{j=0}^{7}|3+8j\rangle$ on 6 qubits, the phase (in $[0°, 360°)$) of the amplitude of $|16\rangle$ after the QFT is

(A) 90° **(B)** ✓ 270° **(C)** 135° **(D)** 0°

Answer: (B) Detailed solution

What you need to know. Peak amplitudes are $\frac{1}{\sqrt{r}}\omega^{x_0 y}$.

Step-by-step solution. From the Q.61 calculation, $\beta_{16} = \frac{1}{\sqrt{8}}\,\omega^{3\cdot 16} = \frac{1}{\sqrt{8}}\,e^{2\pi i\, 48/64} = \frac{1}{\sqrt{8}}\,e^{2\pi i \cdot \frac{3}{4}}$. Phase $= \frac{3}{4} \times 360° = 270°$ (the amplitude is $-\frac{i}{\sqrt{8}}$). The offset 3 is encoded in these phases: $\beta_{8k} \propto e^{2\pi i\, 3k/8}$, which is invisible to a single measurement but is what the shift theorem predicts.

Why the other options are wrong.

(A) 90° uses $e^{-2\pi i}$ or $x_0 = 1$.

(C) 135° uses $x_0 y/N$ with the wrong N.

(D) 0° forgets the offset phase.

Remember. The offset x_0 is encoded as a phase ramp over the peaks.

Q.63 Single-correct MCQ · Shift theorem

Let $\mathrm{QFT}\sum_x \alpha_x |x\rangle = \sum_y \beta_y |y\rangle$ on n qubits ($N = 2^n$, $\omega = e^{2\pi i/N}$). For a fixed integer s, $\mathrm{QFT}\sum_x \alpha_x |(x+s) \bmod N\rangle$ equals

(A) $\sum_y \omega^{-sy}\beta_y |y\rangle$ **(B)** $\sum_y \beta_{(y+s) \bmod N} |y\rangle$

(C) $\sum_y \beta_y |(y+s) \bmod N\rangle$ **(D)** ✓ $\sum_y \omega^{sy}\beta_y |y\rangle$

Answer: (D) Detailed solution

What you need to know. Shift theorem: $\mathrm{QFT}|x+s\rangle = \sum_y \omega^{sy}(\omega^{xy}/\sqrt{N})|y\rangle$.

Step-by-step solution. By linearity it is enough to look at one basis state: $\mathrm{QFT}|x+s\rangle = \frac{1}{\sqrt{N}}\sum_y \omega^{(x+s)y}|y\rangle = \sum_y \omega^{sy}\left(\frac{\omega^{xy}}{\sqrt{N}}\right)|y\rangle$ ($\omega^N = 1$ makes "mod N" harmless). Summing with weights α_x gives $\sum_y \omega^{sy}\beta_y|y\rangle$. **Shift theorem:** a shift in x becomes a phase ramp in y, and $|\beta_y|^2$ is unchanged. This is why the random, unknown offset x_0 left after measuring the work register in Shor's algorithm does not matter.

Why the other options are wrong.

(A) ω^{-sy} is the shift theorem for the inverse QFT.

(B) β_{y+s} shifts the output instead of multiplying by a phase.

(C) $|y+s\rangle$ shifts the output labels.

Remember. Shift ↔ phase ramp; magnitudes unchanged.

Q.64 Single-correct MCQ · QPE precision vs. counting qubits

To obtain φ accurate to n bits with probability at least $1-\varepsilon$, phase estimation uses $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon})\rceil$ counting qubits. For $n = 5$ and $\varepsilon = 0.1$, t equals

(A) 7 **(B)** ✓ 8 **(C)** 10 **(D)** 9

Answer: (B) Detailed solution

What you need to know. $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon})\rceil$.

Step-by-step solution. $2 + \frac{1}{2\times 0.1} = 2+5 = 7$ and $\log_2 7 \approx 2.807$, so $\lceil \log_2 7\rceil = 3$ and $t = 5+3 = 8$. The extra 3 qubits make the outcome distribution (whose peak has width about one grid point 2^{-t}, with tails $\propto 1/\delta^2$) concentrated enough that the first 5 bits are right with probability ≥ 0.9.

Why the other options are wrong.

(A) 7 rounds $\log_2 7$ down.

(C) 10 doubles n.

(D) 9 uses $\varepsilon = 0.05$ or adds one extra.

Remember. A few extra qubits buy confidence; they grow like $\log(1/\varepsilon)$.

Q.65 Single-correct MCQ · QPE with exact t-bit phase

Phase estimation with $t = 3$ counting qubits (inverse QFT readout) is run on an eigenstate with $\varphi = \frac{5}{8}$. The probability that the counting register reads 101 is

(A) $\frac{5}{8}$ **(B)** ✓ 1 **(C)** $\frac{1}{8}$ **(D)** $\frac{4}{\pi^2}$

Answer: (B) Detailed solution

What you need to know. If $\varphi = b/2^t$ exactly, the register is QFT $|b\rangle$ and the inverse QFT returns b with certainty.

Step-by-step solution. After the controlled-U^{2^j} gates the counting register holds $\frac{1}{\sqrt{8}}\sum_{k=0}^{7} e^{2\pi i\varphi k}|k\rangle = \frac{1}{\sqrt{8}}\sum_k \omega^{5k}|k\rangle = \text{QFT}\,|5\rangle$ ($\omega = e^{2\pi i/8}$), because $\varphi = \frac{5}{8} = 0.101_2$ is exactly representable with 3 bits. The inverse QFT returns $|5\rangle = |101\rangle$ with certainty. The bound $\frac{4}{\pi^2}$ applies only when φ is *not* a multiple of 2^{-t}.

Why the other options are wrong.

(A) $\frac{5}{8}$ confuses the phase value with a probability.

(C) $\frac{1}{8}$ assumes a uniform output.

(D) $\frac{4}{\pi^2}$ is the bound for *non*-exact phases.

Remember. Exact binary phases are read perfectly.

Q.66 Single-correct MCQ · The $4/\pi^2$ success bound

If φ is not exactly a t-bit binary fraction, phase estimation returns the *single* closest t-bit approximation with probability at least

(A) $\frac{1}{2}$ **(B)** $\frac{1}{\pi} \approx 0.318$ **(C)** ✓ $\frac{4}{\pi^2} \approx 0.405$ **(D)** $\frac{8}{\pi^2} \approx 0.81$

Answer: (C) Detailed solution

What you need to know. $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ gives $|\alpha_b| \geq 2/\pi$.

Step-by-step solution. Let b be the closest grid point and $\delta = \varphi - b/2^t$, with $0 < |\delta| \leq 2^{-(t+1)}$. The amplitude of b is

$$\alpha_b = \frac{1}{2^t}\sum_{k=0}^{2^t-1} e^{2\pi i k\delta}, \qquad |\alpha_b| = \frac{|\sin(\pi 2^t\delta)|}{2^t|\sin(\pi\delta)|}.$$

Since $\pi 2^t|\delta| \leq \frac{\pi}{2}$, use $\sin x \geq \frac{2x}{\pi}$ there (concavity) in the numerator and $\sin x \leq x$ in the denominator: $|\alpha_b| \geq \frac{2\cdot 2^t|\delta|}{2^t\pi|\delta|} = \frac{2}{\pi}$, so $P(b) \geq \frac{4}{\pi^2} \approx 0.405$. ($\frac{8}{\pi^2}$ is the bound for getting *one of the two* nearest grid points.)

Why the other options are wrong.

(A) $\frac{1}{2}$ is not the proven bound (it can be lower: 0.405).

(B) $\frac{1}{\pi}$ is not a bound from the derivation.

(D) $\frac{8}{\pi^2}$ is for one of the *two* closest points.

Remember. Worst case: φ exactly midway between grid points.

Q.67 Single-correct MCQ · QPE distribution for $\varphi = 1/3$

Phase estimation with $t = 2$ counting qubits is run with $\varphi = \frac{1}{3}$. The probability of the outcome $y = 1$ (i.e. the estimate $\varphi \approx \frac{1}{4}$) is

(A) $\dfrac{6-3\sqrt{3}}{16}$ **(B)** $\dfrac{1}{3}$ **(C)** $\dfrac{4}{\pi^2}$ **(D)** ✓ $\dfrac{6+3\sqrt{3}}{16}$

Answer: (D) Detailed solution

What you need to know. $P(y) = \frac{\sin^2(\pi 2^t \delta)}{2^{2t}\sin^2(\pi\delta)}$, $\delta = \varphi - y/2^t$.

Step-by-step solution. $P(y) = \frac{1}{16}\left|\sum_{k=0}^{3} e^{2\pi i k\delta}\right|^2 = \frac{1}{16}\frac{\sin^2(4\pi\delta)}{\sin^2(\pi\delta)}$ with $\delta = \varphi - \frac{y}{4}$. For $y = 1$: $\delta = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$, so

$$P(1) = \frac{1}{16}\frac{\sin^2(\pi/3)}{\sin^2(\pi/12)} = \frac{1}{16}\cdot\frac{3/4}{(1-\cos\frac{\pi}{6})/2} = \frac{1}{16}\cdot\frac{3}{2-\sqrt{3}} = \frac{3(2+\sqrt{3})}{16} = \frac{6+3\sqrt{3}}{16} \approx 0.700.$$

For completeness $P(0) = \frac{1}{16}$, $P(2) = \frac{3}{16}$, $P(3) \approx 0.050$; the four add to 1. It exceeds the guaranteed $\frac{4}{\pi^2} \approx 0.405$ because $\frac{1}{3}$ is not at the worst position (midway between grid points).

Why the other options are wrong.

(A) $\frac{6-3\sqrt{3}}{16}$ rationalises with the wrong sign ($\frac{3}{2-\sqrt{3}} = 3(2+\sqrt{3})$).

(B) $\frac{1}{3}$ confuses the phase with a probability.

(C) $\frac{4}{\pi^2}$ is only a lower bound.

Remember. Use $\sin^2 a = \frac{1-\cos 2a}{2}$ for exact surds.

Q.68 Single-correct MCQ · Qubit budget of Shor's algorithm

For Shor's algorithm with a work register of $n = \lceil \log_2 N \rceil$ qubits and a counting register of $2n$ qubits, the total number of qubits needed to factor $N = 21$ is

(A) ✓ 15 **(B)** 12 **(C)** 21 **(D)** 10

Answer: (A) Detailed solution

What you need to know. $n = \lceil \log_2 N \rceil$, total $2n + n = 3n$.

Step-by-step solution. $\log_2 21 \approx 4.39$, so $n = 5$ (5 bits hold the residues $0, \ldots, 20$). Counting register $2n = 10$ qubits, work register 5 qubits: total 15. The $2n$ counting qubits make $M = 2^{2n} \geq N^2$, which guarantees that each peak y/M is within $\frac{1}{2r^2}$ of some k/r, so continued fractions can recover r. (The passage uses exactly these $10 + 5$ qubits.)

Why the other options are wrong.

(B) 12 uses $n = 4$ (too small: $2^4 = 16 < 21$).

(C) 21 uses N itself.

(D) 10 is the counting register only.

Remember. $M = 2^{2n} \geq N^2$ guarantees continued fractions work.

Q.69 SINGLE-CORRECT MCQ · Order finding, $a = 7$, $N = 15$

Order finding for $a = 7$, $N = 15$ uses a counting register of $t = 8$ qubits ($M = 256$). The set of possible measurement outcomes y is

(A) $\{0, 128\}$ **(B)** ✓ $\{0, 64, 128, 192\}$ **(C)** $\{0, 4, 8, 12\}$ **(D)** $\{0, 85, 171\}$

Answer: (B) **Detailed solution**

What you need to know. $r \mid M \Rightarrow$ outcomes exactly kM/r.
Step-by-step solution. Powers of 7 mod 15: $7, 49 \equiv 4, 28 \equiv 13, 91 \equiv 1$, so $r = 4$. After measuring (or ignoring) the work register, the counting register is periodic with period 4: $\frac{1}{8}\sum_{j=0}^{63} |x_0 + 4j\rangle$. Since $4 \mid 256$, by Q.61 the outcomes are exactly the multiples of $M/r = 64$: $y \in \{0, 64, 128, 192\}$, each with probability $\frac{1}{4}$, with no leakage. ($\{0, 85, 171\}$ would correspond to $r = 3$.)
Why the other options are wrong.
(A) $\{0, 128\}$ is for $r = 2$.
(C) $\{0, 4, 8, 12\}$ uses $M = 16$.
(D) $\{0, 85, 171\}$ is for $r = 3$.
Remember. Find r classically to predict the outcomes.

Q.70 SINGLE-CORRECT MCQ · Failure mode $\gcd(k, r) > 1$

In the setting of the previous question (order finding for $a = 7$, $N = 15$, $M = 256$), the continued-fraction step turns $y/256$ into k/r'. Which outcome leads to the *wrong* candidate $r' = 2$?

(A) $y = 64$ **(B)** ✓ $y = 128$ **(C)** $y = 0$ **(D)** $y = 192$

Answer: (B) **Detailed solution**

What you need to know. $\gcd(k, r) > 1$ reduces k/r, giving only a divisor of r.
Step-by-step solution. $\frac{64}{256} = \frac{1}{4} \Rightarrow r' = 4$ ✓; $\frac{192}{256} = \frac{3}{4} \Rightarrow r' = 4$ ✓; $\frac{128}{256} = \frac{1}{2} \Rightarrow r' = 2$; $\frac{0}{256} = \frac{0}{1} \Rightarrow r' = 1$ (no information). For $y = 128$, $k = 2$ shares the factor 2 with $r = 4$, so the fraction reduces and only a *divisor* of r appears. The check $7^2 = 49 \equiv 4 \not\equiv 1 \pmod{15}$ rejects it; one repeats the run or takes the lcm of several candidates. With the correct $r = 4$: $\gcd(7^2 - 1, 15) = \gcd(48, 15) = 3$, $\gcd(7^2 + 1, 15) = \gcd(50, 15) = 5$.
Why the other options are wrong.
(A) $64/256 = \frac{1}{4}$ gives $r = 4$ correctly.
(C) 0 gives $\frac{0}{1}$, no information ($r' = 1$), not 2.
(D) $192/256 = \frac{3}{4}$ gives $r = 4$ correctly.
Remember. Check $a^{r'} \equiv 1$; take the lcm of candidates.

Q.71 SINGLE-CORRECT MCQ · Approximate QFT gate savings

An approximate QFT on $n = 20$ qubits keeps only the controlled-R_k gates with $k \le m = 6$ and drops the rest. The number of controlled rotations dropped is

(A) 114 **(B)** 85 **(C)** 95 **(D)** ✓ 105

Answer: (D) **Detailed solution**

What you need to know. R_k appears $n - k + 1$ times; total $\binom{n}{2}$.
Step-by-step solution. In the textbook circuit the gate R_k (for $k = 2, \ldots, n$) appears $n - k + 1$ times (once for each target qubit j with $j + k - 1 \le n$). Total $= \binom{20}{2} = 190$. Kept: $k = 2, \ldots, 6$: $19 + 18 + 17 + 16 + 15 = 85$. Dropped: $190 - 85 = 105$. The approximate QFT therefore needs only $O(nm)$ gates instead of $O(n^2)$, at the price of a small error (Q.77).

Why the other options are wrong.

(A) 114 is not a count from the circuit.

(B) 85 is the number *kept*.

(C) 95 is not a count from the circuit.

Remember. Approximate QFT needs $O(nm)$ gates.

Q.72 SINGLE-CORRECT MCQ · Periodic state with $r \nmid N$

On $n = 6$ qubits ($N = 64$) the uniform superposition of all $x \equiv 0 \pmod 6$, $0 \le x \le 63$, is prepared and the QFT applied. The probability of the outcome $y = 32$ is

(A) ✓ $\frac{11}{64}$ **(B)** $\frac{1}{64}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{1}{11}$

Answer: (A) Detailed solution

What you need to know. At y with ry/N an integer all phases are 1: $P = K^2/(KN) = K/N$.

Step-by-step solution. The allowed x are $0, 6, 12, \ldots, 60$: $K = 11$ terms ($6 \times 10 = 60 \le 63 < 66$). Then

$$\beta_{32} = \frac{1}{\sqrt{11}} \cdot \frac{1}{\sqrt{64}} \sum_{j=0}^{10} e^{2\pi i\, 6j \cdot 32/64} = \frac{1}{\sqrt{704}} \sum_{j=0}^{10} e^{2\pi i \cdot 3j} = \frac{11}{\sqrt{704}},$$

so $P(32) = \frac{121}{704} = \frac{11}{64} \approx 0.172$. The period 6 does not divide 64, so the ideal answer $\frac{1}{6}$ is only approximate, and most peaks ($y \approx 10.67k$) are smeared over neighbouring integers; $y = 32 = 3 \cdot \frac{64}{6}$ happens to fall exactly on a grid point.

Why the other options are wrong.

(B) $\frac{1}{64}$ assumes a flat distribution.

(C) $\frac{1}{6}$ is the idealised value when $r \mid N$.

(D) $\frac{1}{11}$ is the probability per input term, not of the outcome.

Remember. Count terms exactly when $r \nmid N$.

Q.73 SINGLE-CORRECT MCQ · Eigenvalues of the QFT

Every eigenvalue λ of the n-qubit QFT matrix ($n \ge 1$) satisfies

(A) $\lambda^N = 1$, and all N of the N-th roots of unity occur

(B) $|\lambda| < 1$ for at least one eigenvalue

(C) $\lambda \in \{1, -1\}$

(D) ✓ $\lambda \in \{1, -1, i, -i\}$

Answer: (D) Detailed solution

What you need to know. $\mathrm{QFT}^4 = I \Rightarrow \lambda^4 = 1$.

Step-by-step solution. From Q.48, $\mathrm{QFT}^2 |x\rangle = |-x\rangle$, hence $\mathrm{QFT}^4 = I$. If $F\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v} = F^4\mathbf{v} = \lambda^4\mathbf{v}$, so $\lambda^4 = 1$: $\lambda \in \{1, -1, i, -i\}$. The QFT is unitary, so $|\lambda| = 1$ always. Imaginary eigenvalues do occur (e.g. i for $N = 4$, where $\operatorname{tr} F = 1 + i$), so $\{1, -1\}$ is too small; for $N \ge 8$ all four values occur, but never all N roots of unity once $N > 4$.

Why the other options are wrong.

(A) Not all N-th roots occur; only the four 4th roots.

(B) $|\lambda| = 1$ for every eigenvalue of a unitary.

(C) $\{1, -1\}$ is too small: i occurs already for $N = 4$.

Remember. Spectrum of the Fourier transform: $\{\pm 1, \pm i\}$.

Q.74 SINGLE-CORRECT MCQ · Semiclassical QFT

In phase estimation the inverse QFT is immediately followed by measurement of every counting qubit. Using the semiclassical (Griffiths–Niu) QFT, the number of *two-qubit* gates needed for this inverse QFT on n qubits is

(A) $n-1$ **(B)** $\dfrac{n(n-1)}{2}$ **(C)** ✓ 0 **(D)** n

Answer: (C) Detailed solution

What you need to know. Measure-and-feed-forward replaces each controlled rotation by a classically controlled single-qubit rotation.

Step-by-step solution. Controlled-phase gates are symmetric (Q.27), so in each $CR_k^\dagger$ the qubit that is measured *first* can be taken as the control. Once a qubit has been measured, its value y_j is a classical bit, and "controlled on a measured qubit" becomes "apply $R_k^\dagger$ if $y_j = 1$": a classically controlled single-qubit gate. Measuring the qubits one after the other and feeding each result forward removes all $\frac{n(n-1)}{2}$ two-qubit gates. This needs only mid-circuit measurement plus feed-forward, and even allows one counting qubit to be recycled n times.

Why the other options are wrong.

(A) $n-1$ would keep one gate per qubit; none are needed.

(B) $\binom{n}{2}$ is the fully quantum circuit.

(D) n confuses gates with measurements.

Remember. Symmetric controlled phases + early measurement = no two-qubit gates.

Q.75 MATCHING LISTS · Applications of the QFT

Match each algorithm in List-I with the role played by the QFT, or the related step, in List-II.

List-I		List-II	
(P)	Quantum phase estimation	**(1)**	adds a number by applying controlled phase rotations to a register held in the Fourier basis
(Q)	Shor's factoring algorithm	**(2)**	turns the phases $e^{2\pi i\, 2^j \varphi}$ in the counting register into the binary digits of φ
(R)	Draper's quantum adder	**(3)**	after a QPE step, applies an ancilla rotation proportional to $1/\lambda$ for each eigenvalue λ
(S)	HHL linear-system solver	**(4)**	extracts an order r from peaks near kM/r using continued fractions

(A) ✓ P→2, Q→4, R→1, S→3 **(B)** P→4, Q→2, R→1, S→3

(C) P→2, Q→1, R→4, S→3 **(D)** P→2, Q→4, R→3, S→1

Answer: (A) Detailed solution

What you need to know. Know the role of the QFT in each algorithm.

Step-by-step solution. QPE: controlled-U^{2^j} writes $\frac{1}{\sqrt{2^t}}\sum_k e^{2\pi i\varphi k}\,|k\rangle \approx \mathrm{QFT}\,|2^t\varphi\rangle$; the inverse QFT reads the digits (2). Shor: QPE applied to modular multiplication, whose eigenphases are k/r; continued fractions give r (4). Draper: in the Fourier basis, adding a to $|x\rangle$ is just the phase ramp ω^{ay} (Q.63), done with single-qubit or controlled rotations, then an inverse QFT (1). HHL: QPE puts eigenvalues λ_j in a register, a controlled rotation of an ancilla by an angle with $\sin\theta \propto 1/\lambda_j$ performs A^{-1}, then QPE is uncomputed (3). Hence P→2, Q→4, R→1, S→3.

Why the other options are wrong.

(B) P→4, Q→2 swaps QPE and Shor.

(C) Q→1, R→4 swaps Shor and Draper.

(D) R→3, S→1 swaps Draper and HHL.

Remember. QPE reads phases; Shor = QPE + continued fractions; Draper adds in the Fourier basis; HHL inverts eigenvalues.

Q.76 MATCHING LISTS · Period ↔ peak spacing

On $n = 4$ qubits ($N = 16$) the QFT is applied to each input in List-I and the result measured. Match with List-II.

List-I		List-II	
(P)	$\|0\rangle$	**(1)**	only $y = 0$ and $y = 8$, each with probability $\frac{1}{2}$
(Q)	$\frac{1}{2}(\|0\rangle + \|4\rangle + \|8\rangle + \|12\rangle)$	**(2)**	all 16 outcomes, each with probability $\frac{1}{16}$
(R)	$\frac{1}{\sqrt{8}}\sum_{x \text{ even}} \|x\rangle$	**(3)**	only $y = 0, 4, 8, 12$, each with probability $\frac{1}{4}$
(S)	$\frac{1}{4}\sum_{x=0}^{15} \|x\rangle$	**(4)**	$y = 0$ with certainty

(A) P→4, Q→3, R→1, S→2

(B) P→2, Q→3, R→4, S→1

(C) ✓ P→2, Q→3, R→1, S→4

(D) P→2, Q→1, R→3, S→4

Answer: (C) Detailed solution

What you need to know. Period $r \mid 16 \Rightarrow$ peaks every $16/r$.

Step-by-step solution. P: a basis state transforms to equal-magnitude amplitudes (Q.42): uniform over 16 (2). Q: period $r = 4$ dividing 16: peaks at multiples of $16/4 = 4$, i.e. $0, 4, 8, 12$, each $\frac{1}{4}$ (3). R: the even numbers have period $r = 2$: peaks at multiples of $16/2 = 8$, i.e. 0 and 8, each $\frac{1}{2}$ (1). S: the uniform superposition has period 1 and goes to $|0\rangle$ (Q.49) (4). Pattern: period r in $x \Leftrightarrow r$ peaks spaced N/r in y — the "uncertainty principle" of the Fourier transform. Hence P→2, Q→3, R→1, S→4.

Why the other options are wrong.

(A) P→4 says QFT $|0\rangle = |0\rangle$; it is uniform.

(B) R→4, S→1 swaps even-x and uniform states.

(D) Q→1, R→3 swaps period 4 and period 2.

Remember. Larger period ⇒ more, closer peaks.

Q.77 ASSERTION–REASON · Approximate QFT error bound

Assertion (A): For large n, dropping every controlled-R_k with $k > m$, where m is only slightly larger than $\log_2 n$, changes the output of the QFT circuit only slightly.

Reason (R): Each dropped gate differs from the identity only by a phase $2\pi/2^k$, so its error is at most about $2\pi/2^k$; at most n gates are dropped for each k, giving a total error of order $n \cdot 2\pi/2^m$, which is small once $2^m \gg n$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

What you need to know. $\|CR_k - I\| \le 2\pi/2^k$; errors of a product add.
Step-by-step solution. **R is true:** $\|CR_k - I\| = |e^{2\pi i/2^k} - 1| = 2\sin(\pi/2^k) \le 2\pi/2^k$. Errors of a product of unitaries add at most linearly (triangle inequality), and R_k appears $n-k+1 \le n$ times, so the total error is at most $\sum_{k>m} n \cdot \frac{2\pi}{2^k} < \frac{2\pi n}{2^m}$. **A is true** *because* of R: choosing $m = \log_2 n + c$ makes the error at most $2\pi/2^c$, independent of n. (Coppersmith's approximate QFT; in practice gate noise is larger than the error from these tiny rotations, so dropping them even *improves* real results.) Answer: (A).
Why the other options are wrong.
(B) needs A and R both true but R not explaining A; R is exactly the error estimate behind A.
(C) needs A true and R false; R is true.
(D) needs A false and R true; A is true.
Remember. Tiny rotations can be dropped with bounded error.

Q.78 Statement I / II · Non-terminating binary phases

Statement I: For $\varphi = \frac{1}{3}$, phase estimation with any finite number t of counting qubits never returns an outcome y with $y/2^t = \varphi$ exactly, with certainty or otherwise.

Statement II: The number $\frac{1}{3}$ has no terminating binary expansion ($\frac{1}{3} = 0.010101\ldots_2$).

(A) ✓ Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

What you need to know. Outcomes are integers y; $y/2^t = \frac{1}{3}$ would need $3 \mid 2^t$.
Step-by-step solution. **II is true:** $\frac{1}{3} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots = 0.\overline{01}_2$; a terminating expansion would mean $\frac{1}{3} = \frac{y}{2^t}$, i.e. $2^t = 3y$, impossible since $3 \nmid 2^t$. **I is true:** the outcome is always an integer y, so the estimate is $y/2^t \ne \frac{1}{3}$ for every t by the same argument. The output distribution is peaked near $2^t/3$ (see Q.67: $t=2$ gives $y=1$ with probability ≈ 0.70). Both statements are true. Answer: (A).
Why the other options are wrong.
(B) needs both statements false; both are true.
(C) needs I true and II false; II is true.
(D) needs I false and II true; I is true.
Remember. Non-dyadic phases are only approximated.

Paragraph for Questions 79 and 80

Shor's algorithm for $N = 21$ with $a = 2$. A counting register of $t = 10$ qubits ($M = 2^{10} = 1024$) and a 5-qubit work register initialised to $|1\rangle$ are used. Steps: (i) $H^{\otimes 10}$ on the counting register; (ii) $|x\rangle\,|1\rangle \mapsto |x\rangle\,|2^x \bmod 21\rangle$; (iii) inverse QFT on the counting register; (iv) measure the counting register, obtaining y. Classical post-processing replaces $y/1024$ by the closest fraction k/r with $r < 21$ (continued fractions) and keeps r if $2^r \equiv 1 \pmod{21}$. The true order of 2 modulo 21 is 6, since $2^6 = 64 = 3 \cdot 21 + 1$.

Q.79 PARAGRAPH BASED · Shor, $N = 21$: continued fractions

Which of the following measured values y leads the post-processing to the correct order $r = 6$?

(A) 683 **(B)** 512 **(C)** ✓ 171 **(D)** 341

Answer: (C) Detailed solution

What you need to know. Continued fractions of y/M; good outcomes have $\gcd(k, r) = 1$.
Step-by-step solution. The peaks lie near $y \approx k \cdot \frac{1024}{6} = 170.67k$: $0, 170.67, 341.33, 512, 682.67, 853.33$.

- $\frac{171}{1024} = 0.16699$: continued fraction [0; 5, 1, 84, 2], convergents $\frac{1}{5}, \frac{1}{6}, \frac{85}{509}, \frac{171}{1024}$; the last with denominator < 21 is $\frac{1}{6} \Rightarrow r = 6$, and $2^6 \equiv 1$ ✓.
- $\frac{341}{1024} \approx \frac{1}{3}$ ($k = 2$), $\frac{512}{1024} = \frac{1}{2}$ ($k = 3$), $\frac{683}{1024} \approx \frac{2}{3}$ ($k = 4$): $\gcd(k, 6) > 1$, so only the divisors $3, 2, 3$ appear and $2^3 = 8$, $2^2 = 4$ are $\not\equiv 1 \pmod{21}$.

Factoring: $2^{r/2} = 8$, $\gcd(8 - 1, 21) = 7$ and $\gcd(8 + 1, 21) = 3$; indeed $21 = 3 \times 7$. Only $k = 1, 5$ (coprime to 6) give r directly, i.e. about $\frac{\varphi(6)}{6} = \frac{1}{3}$ of the runs.
Why the other options are wrong.
(A) $683 \to \frac{2}{3}$ ($k = 4$): divisor 3.
(B) $512 \to \frac{1}{2}$ ($k = 3$): divisor 2.
(D) $341 \to \frac{1}{3}$ ($k = 2$): divisor 3.
Remember. Only $\varphi(r)/r$ of the peaks give r directly.

Q.80 PARAGRAPH BASED · Shor, $N = 21$: peak probability

The probability that the measured value is exactly $y = 512$ is closest to

(A) 0.333 **(B)** 0.500 **(C)** ✓ 0.167 **(D)** 0.001

Answer: (C) Detailed solution

What you need to know. Group by work-register value; count terms per branch; on an exact grid point all phases are 1.
Step-by-step solution. After step (ii) the state is $\frac{1}{\sqrt{1024}} \sum_{x=0}^{1023} |x\rangle\,|2^x \bmod 21\rangle$. Group $x = x_0 + 6j$ with $x_0 \in \{0, \ldots, 5\}$; because $1024 = 6 \cdot 170 + 4$, the offsets $x_0 = 0, 1, 2, 3$ have $K = 171$ terms and $x_0 = 4, 5$ have $K = 170$ terms (check: $4 \cdot 171 + 2 \cdot 170 = 1024$). The six work values $2^{x_0} \bmod 21$ are distinct, so the branches do not interfere. After the inverse QFT, the amplitude of $|y\rangle$ in branch x_0 is $\frac{1}{1024} \sum_j e^{-2\pi i (x_0 + 6j) y / 1024}$. For $y = 512$, $e^{-2\pi i \cdot 6j \cdot 512/1024} = e^{-6\pi i j} = 1$, so its modulus is $K_{x_0}/1024$ and

$$P(512) = \frac{4 \cdot 171^2 + 2 \cdot 170^2}{1024^2} = \frac{174\,764}{1\,048\,576} \approx 0.16667 \approx \frac{1}{6}.$$

$y = 512 = 3 \cdot \frac{1024}{6}$ falls exactly on a grid point, so this peak is "perfect"; the peaks at $k = 1, 2, 4, 5$ are smeared over neighbouring y (the nearest integers $171, 341, 683, 853$ get only about 0.11 each).
Why the other options are wrong.

(A) 0.333 would double the peak weight.

(B) 0.500 treats $r = 2$.

(D) 0.001 $\approx 1/1024$ assumes a flat distribution.

Remember. $P(\text{exact peak}) = \sum K_{x_0}^2 / M^2 \approx 1/r$.

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